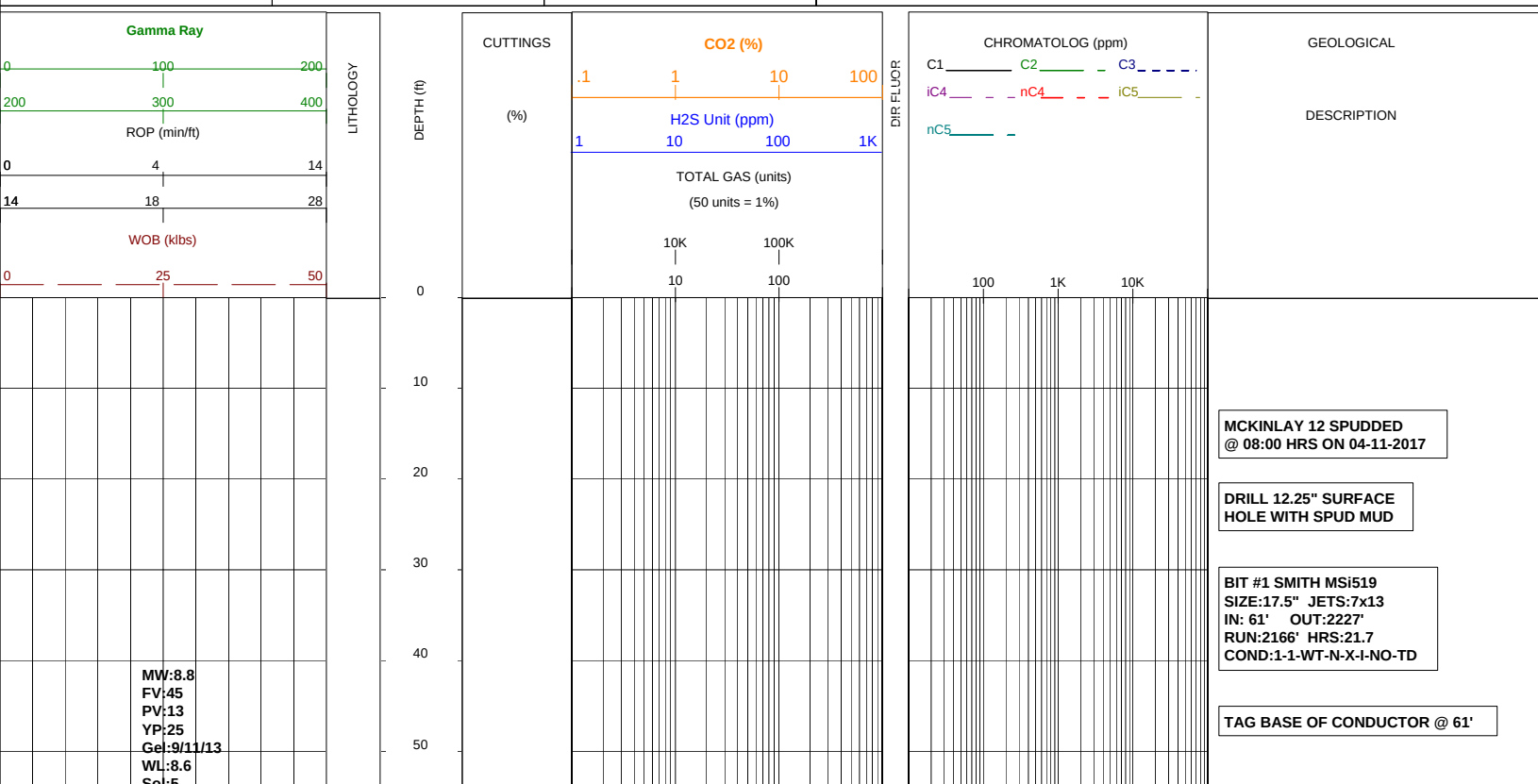
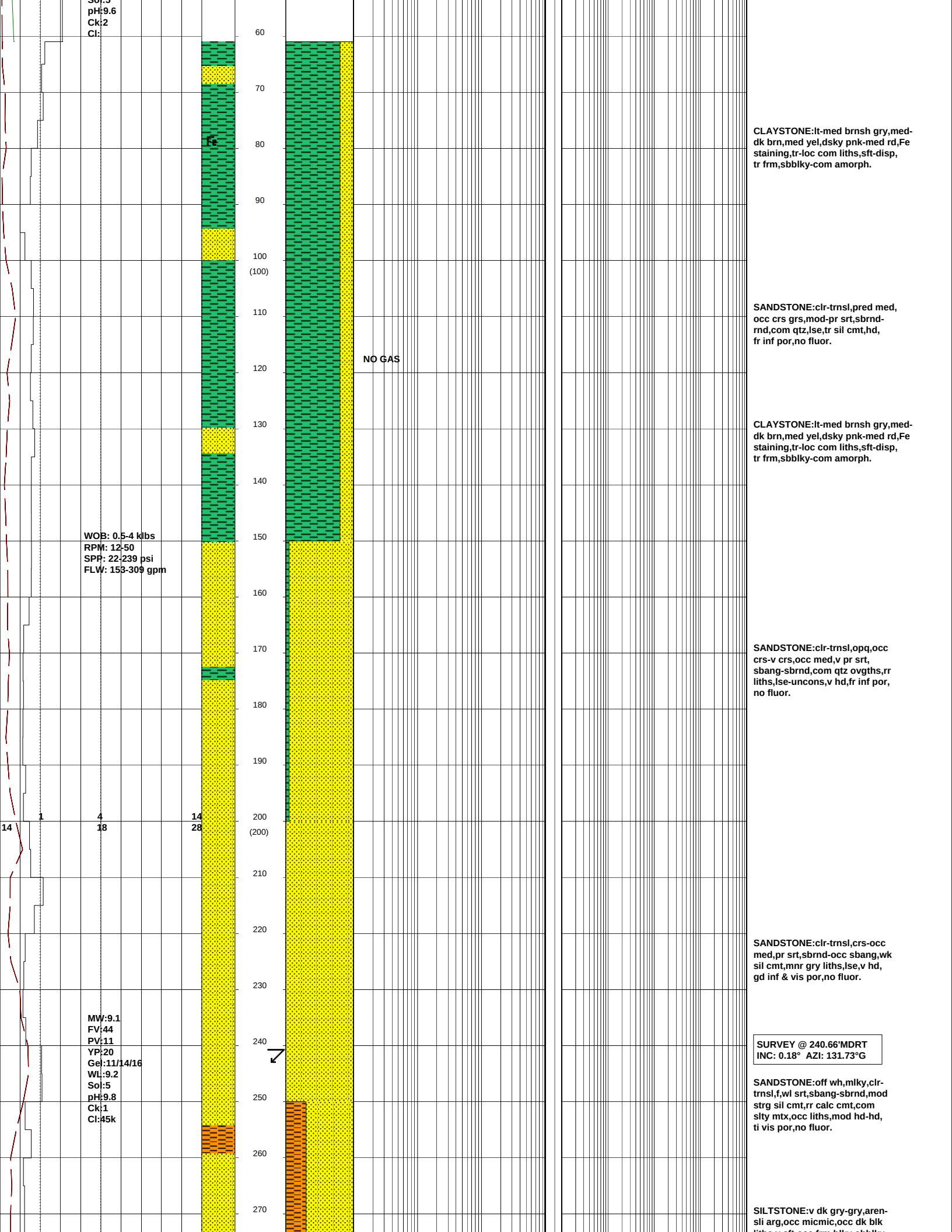
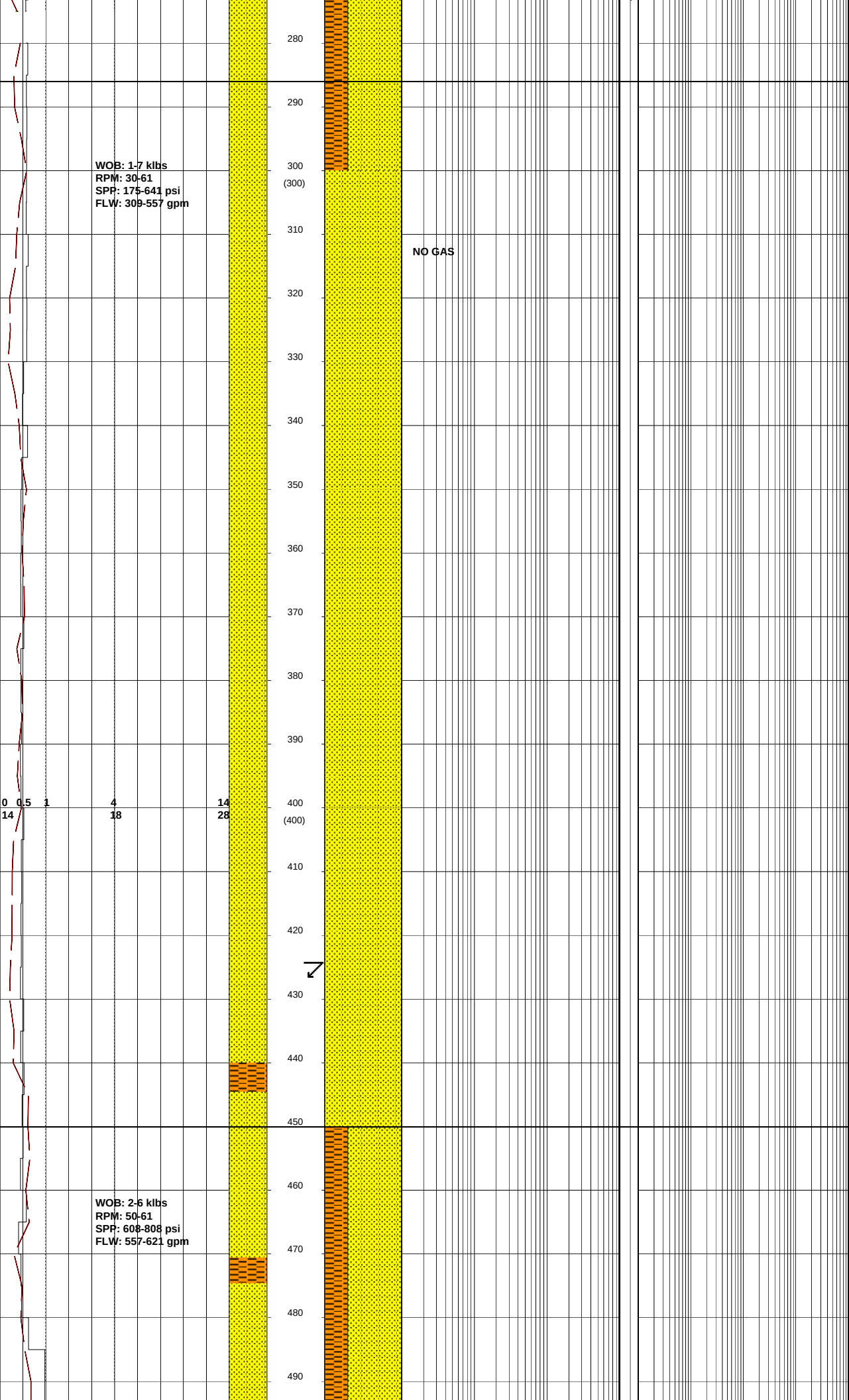


Field : MCKINLAY	Rotary Table : 170.7'	Rig : ENSIGN 971	Open Hole:	Cased Hole:	Loggers : VAHID JALEH
Permit : PPL234	Ground Level : 147.2'	Spud Date: 04-11-2017	17.50" 2227'	13.375" 2217.2"	PHILIP LUU
State : SOUTH AUSTRALIA	GDA 94 Co-ordinates :	TD Date : 17-11-2017	12.25" 5030'	9.625" 5020'	MATHEW TOWNSEND
Country : AUSTRALIA	Lat. : 28°27'53.81"S	Total Depth : 9526'MDRT 4105.9'TVD	8.5" 9526'	5.5" XXXX'	
Scale : 1/240 5":100'	Long. : 140°29'55.95"E	Final Status : TBA			

LITHOLOGY	ACCESSORIES	DRILLING DATA	ABBREVIATIONS
 Coarse sandstone  Medium sandstone  Fine sandstone  Cement  Siltstone  Carb Siltstone  Metamorphic rock  Claystone  Limestone  Coal-lignite	 Pyrite  Siderite  Glauconite  Iron  Mica  Chert  Calcareous  Dolomitic  Carbonaceous  Fossils	 Casing shoe  Bit Trip  Wiper Trip  Core  Dst  Survey MUD DATA MW - Mud Weight (lb/gal) FV - Funnel Viscosity (s/qt) PV - Plastic Viscosity (cps) YP - Yield Point (lb/100ftsq) Gel - Gel Strength (10sec) WL - Water Loss (cc/30min) pH - Acidity / Alkalinity Ck - Cake (32nd/inch) Sol - Solids (% vol) Cl - Chlorides (mg/l)	ABBREVIATIONS BOPD - Barrels of Oil Per Day BWPD - Barrels of Water Per Day CG - Connection Gas CO - Circulate Out COND - Condensate c/c - Crush Cut DST - Drill Stem Test FLOW - Flow Rate (gal/min) GCM - Gas Cut Mud GCW - Gas Cut Water GTS - Gas To Surface INJ - Injection of Mist (bbls/hr) LCM - Lost Circulation Material MMCFD - Million Cubic Feet / Day NGTS - No Gas To Surface NOTS - No Oil To Surface NR - No Returns OCM - Oil Cut Mud OG - Over Gauge OH - Open Hole OTS - Oil To Surface Q - Flow Rate REC - Recovery Rmf - Resistivity mud filtrate ROP - Rate Of Penetration RPM - Revolutions Per Minute RTSTM - Rate Too Small To Measure Rw - Resistivity water r/r - Ring Residue SCFM - Standard Cubic Ft/Min (air) SGCM - Slightly Gas Cut Mud SPM - Strokes Per Minute SPP - Stand Pipe Pressure SWC - Side-Wall Core TG - Trip Gas WOB - Weight On Bit







itns, v sit, occ frm, biky-sbbiky.

EYRE FORMATION:
286.3' MDRT (-115° TVDSS)

SANDSTONE: clr-trns, occ off
wh, crs-v crs, mod wl srt, sbang-
sbrnd, rr wh slty mt, occ qtz
ovgrs, pred lse, v hd grs, gd
inf por, no fluor.

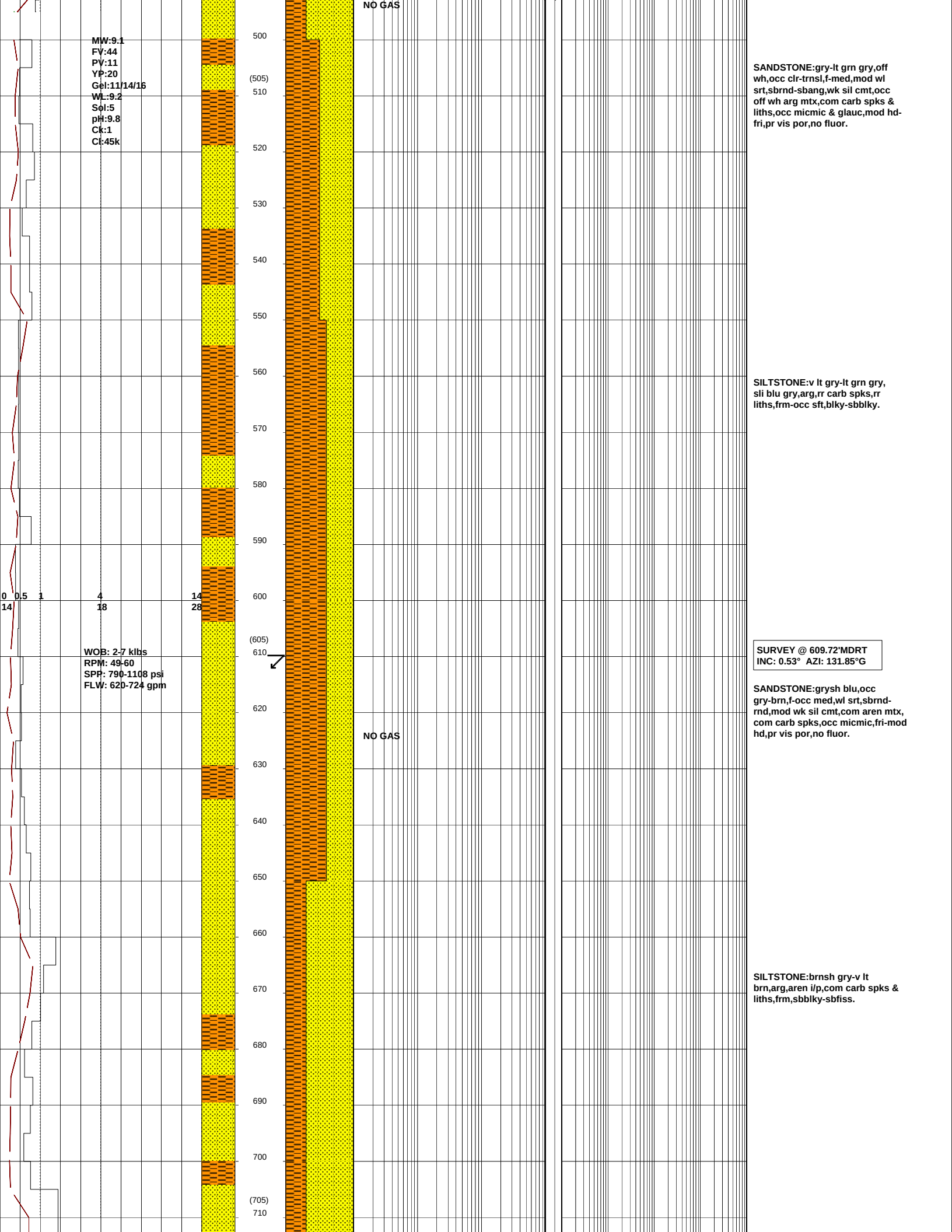
SANDSTONE: clr-trns, occ off
wh, crs-v crs, mod wl srt, sbang-
sbrnd, rr wh slty mt, occ qtz
ovgrs & carb aggs, pred lse, v
hd grs, gd inf por, no fluor.

SANDSTONE: clr-trns, occ off
wh, med-crs, occ v crs, mod wl srt,
sbang-sbrnd, rr wh slty mt, occ qtz
ovgrs & carb spks, pred lse, v
hd grs, gd inf por, no fluor.

SURVEY @ 424.20' MDRT
INC: 0.70° AZI: 143.56° G

WINTON FORMATION:
450' MDRT (-279° TVDSS)

SILTSTONE: med gry-dk gry,
arg, com micmic, com carb spks,
rr glauc, frm, sbblky.



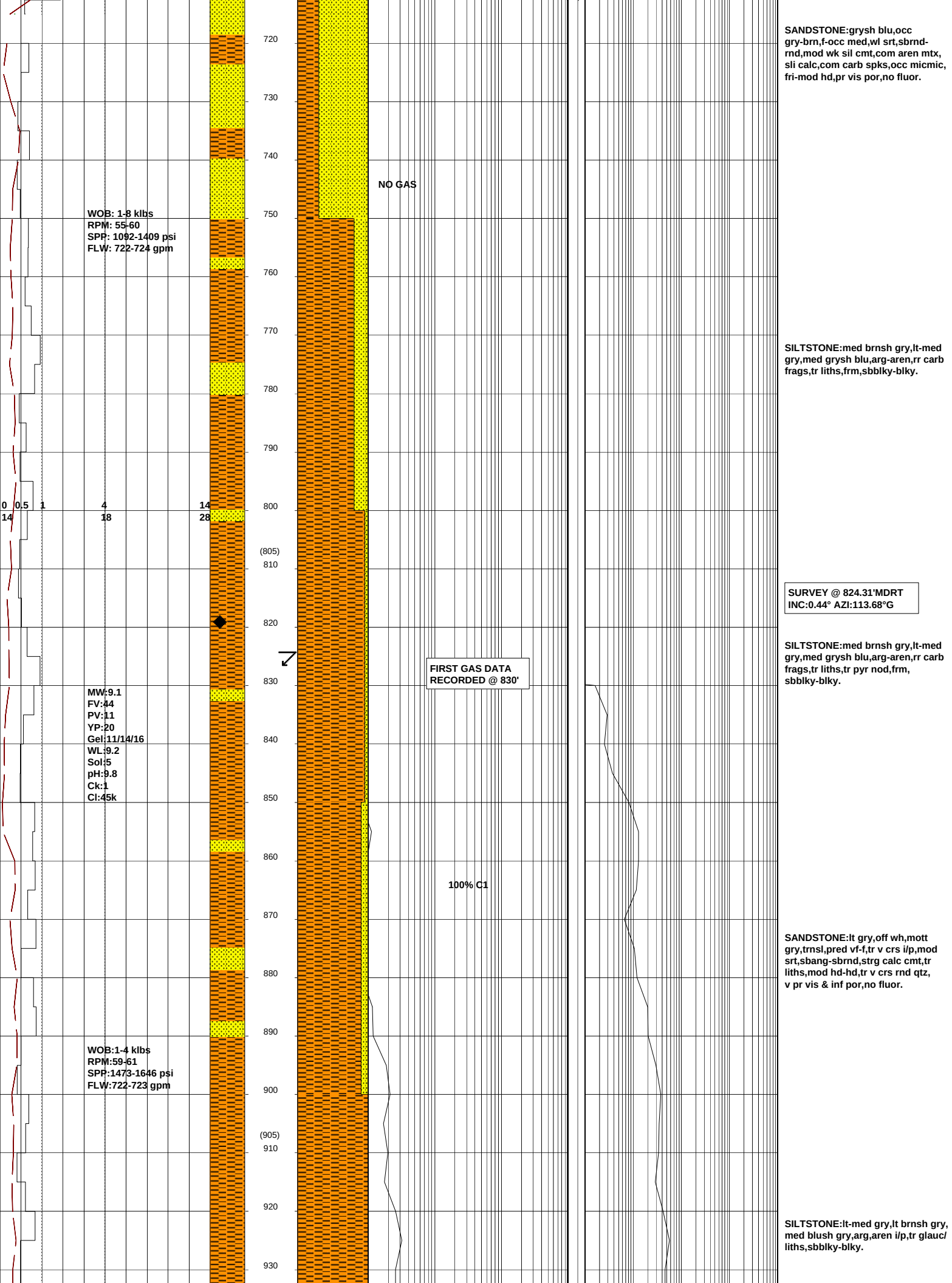
MW:9.1
FV:44
PV:11
YP:20
Gel:11/14/16
WL:9.2
Sol:5
pH:9.8
Ck:1
Ct:45k

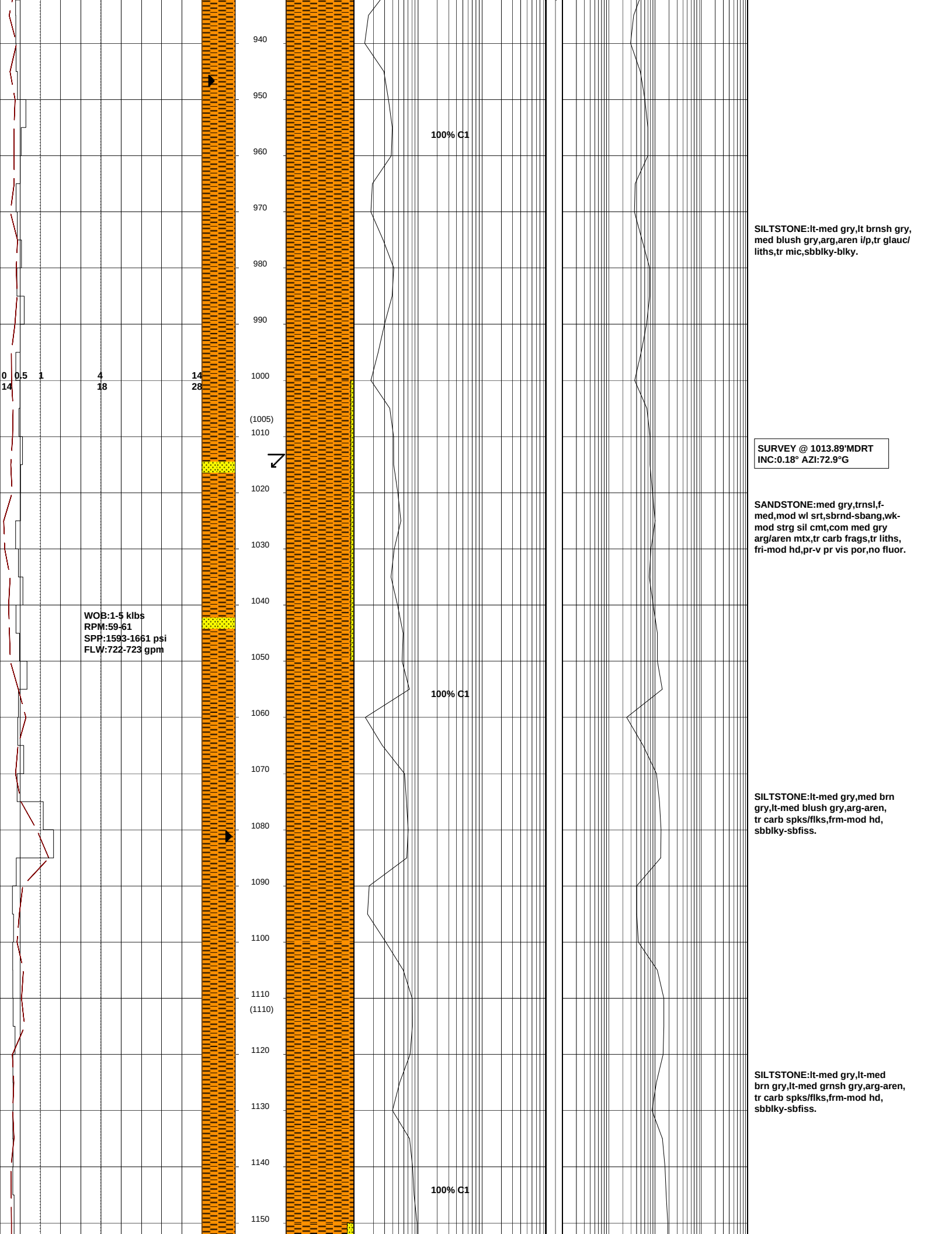
WOB: 2.7 klbs
RPM: 49-60
SPR: 790-1108 psi
FLW: 620-724 gpm

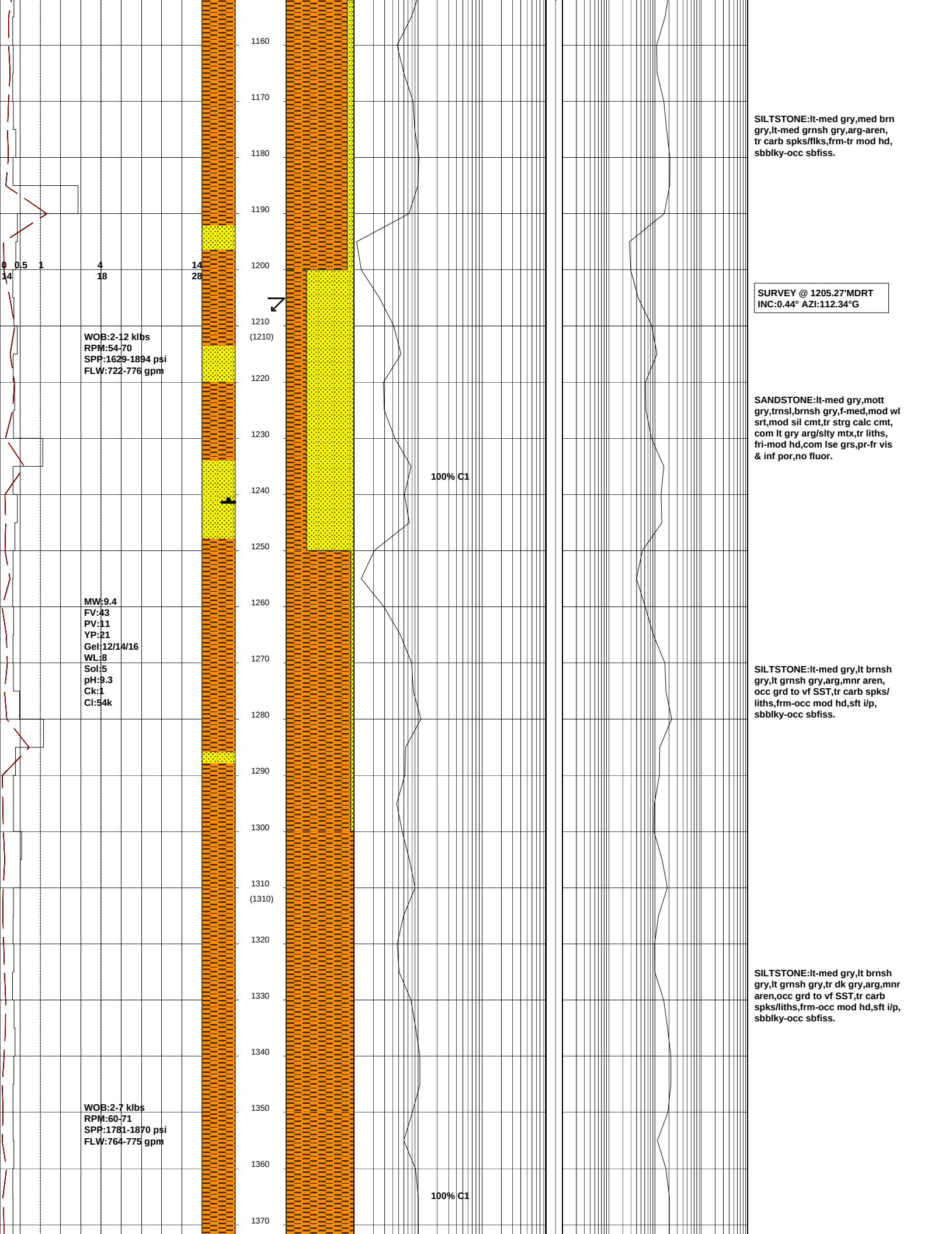
SURVEY @ 609.72' MDRT
INC: 0.53° AZI: 131.85°G

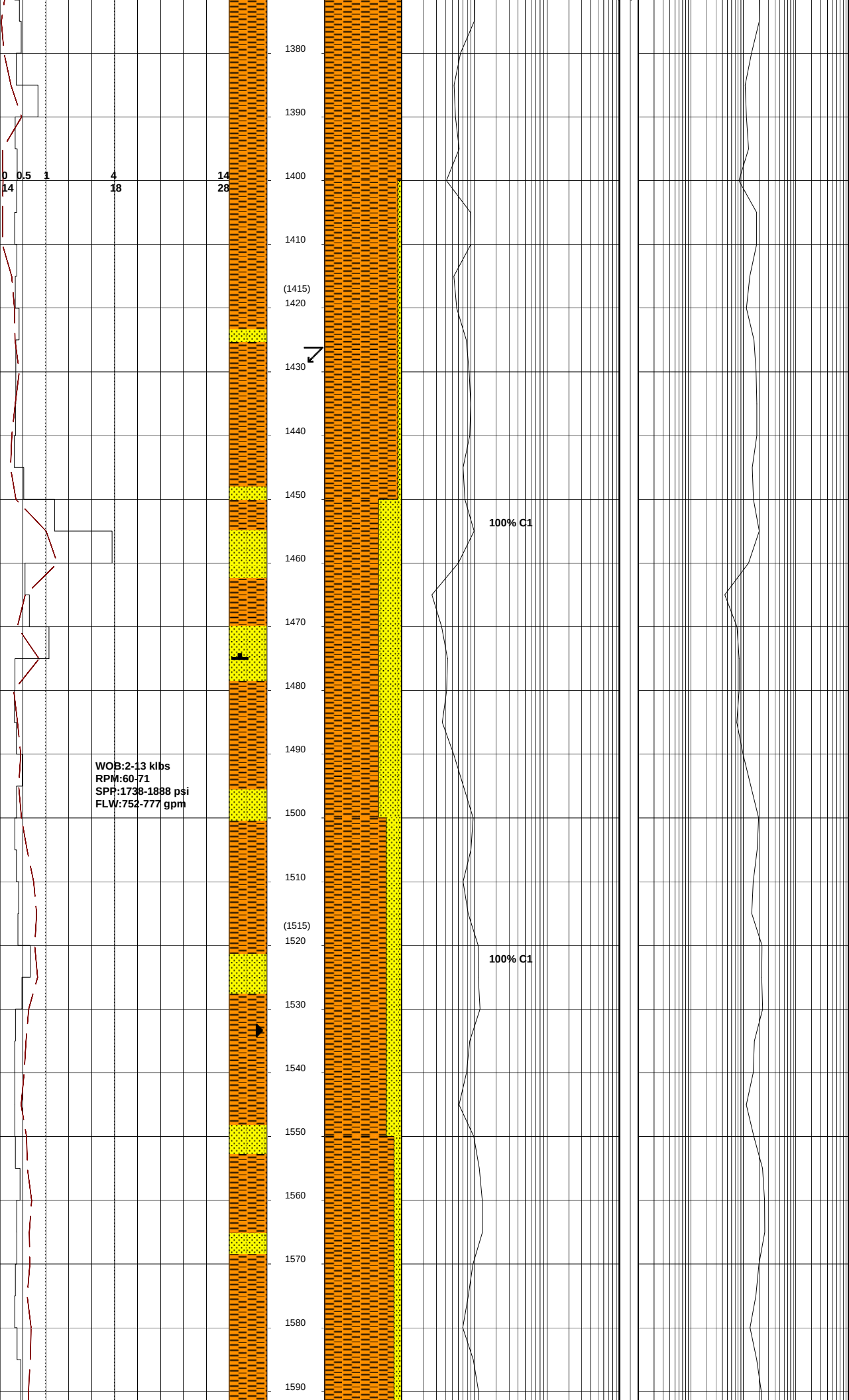
SANDSTONE: grysh blu, occ gry-brn, f-occ med, wl srt, sbrnd-rnd, mod wk sil cmt, com aren mtb, com carb spks, occ micmic, fri-mod hd, pr vis por, no fluor.

SILTSTONE: brnsh gry-v lt brn, arg, aren i/p, com carb spks & liths, frm, sbbly-sbfiss.









SILTSTONE:lt-med gry,lt brnsh gry,lt grnsh gry,tr dk gry,arg,mnr aren,occ grd to vf SST,tr carb spks/liths,frm-occ mod hd,sft i/p,sbblky-occ sbfiss.

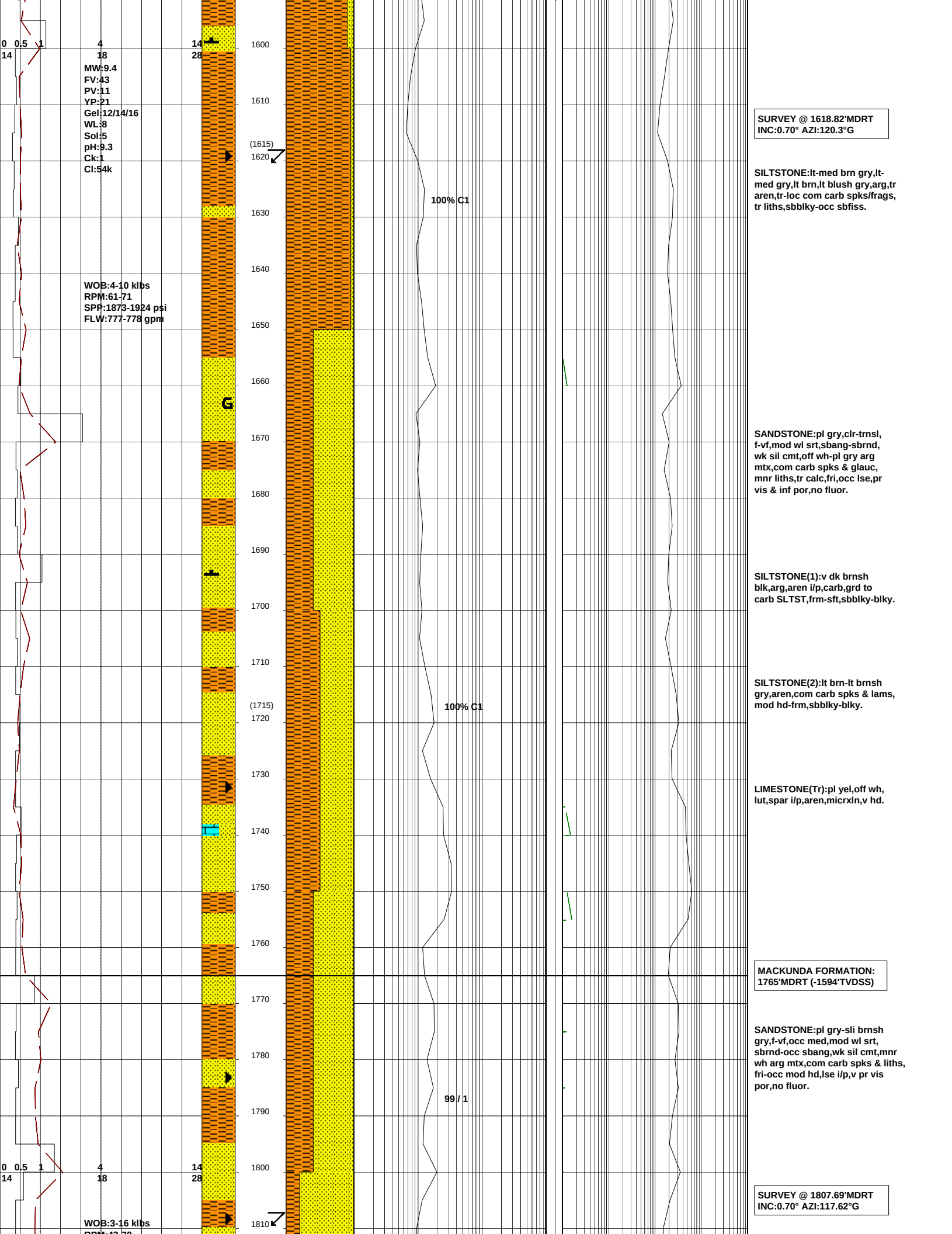
SURVEY @ 1426.6' MDRT
INC:0.44° AZI:94.78°G

SANDSTONE:lt gry,off wh,trnsl, f-med,mod srt,sbang-sbrnd,strg calc cmt,tr wh arg mtx,tr glauc/ liths,hd,v pr-ti vis por,no fluor.

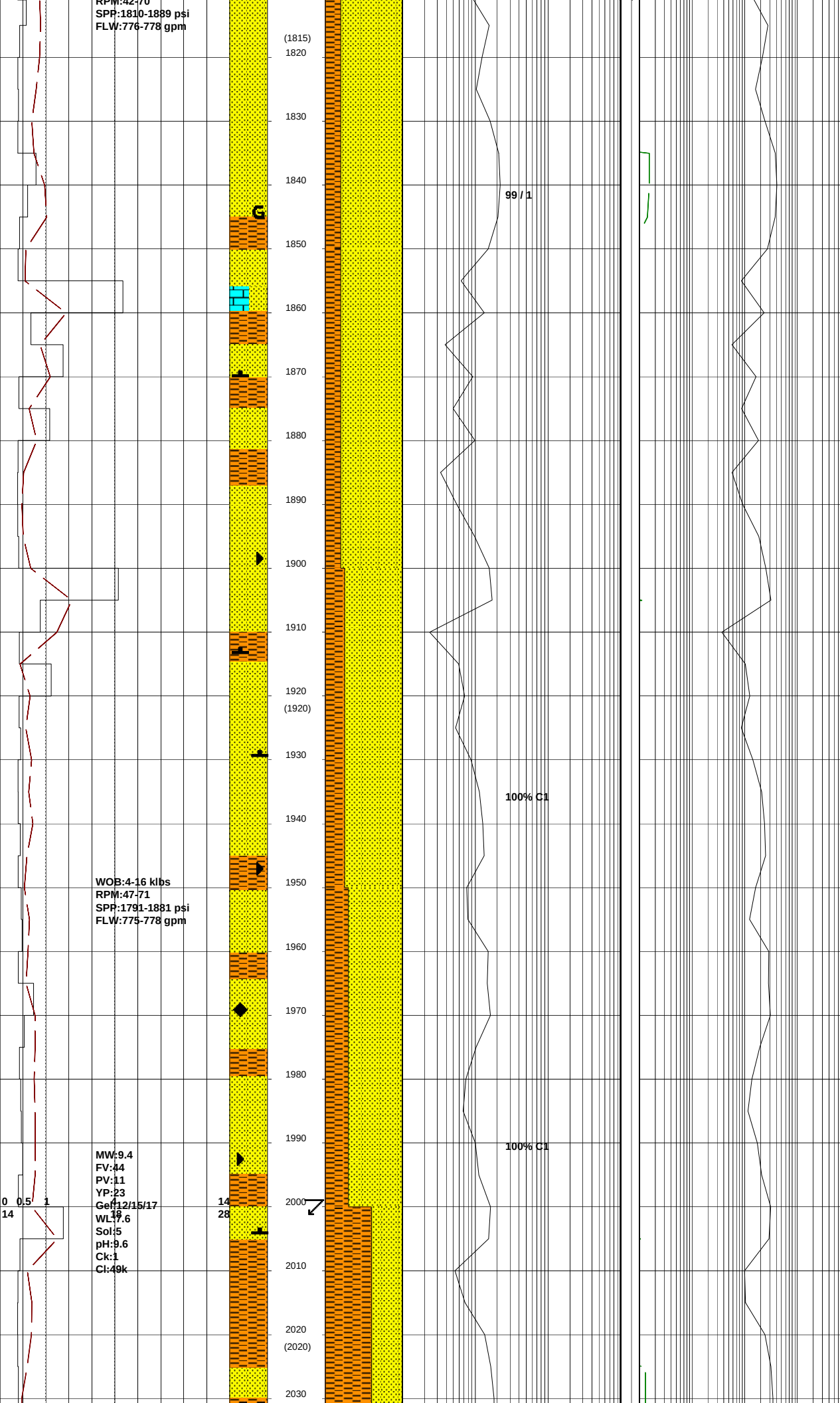
SANDSTONE:lt gry,off wh,mott gry,vf-f,mod wl srt,sbang-sbrnd, mod sil cmt,occ strg calc cmt, com lt gry arg mtx,tr liths,tr glauc, fri aggs,pr-fr vis por,no fluor.

SILTSTONE:lt-med brnsh gry, med gry,arg,tr aren,tr COAL frags,frm,sbblky-blky.

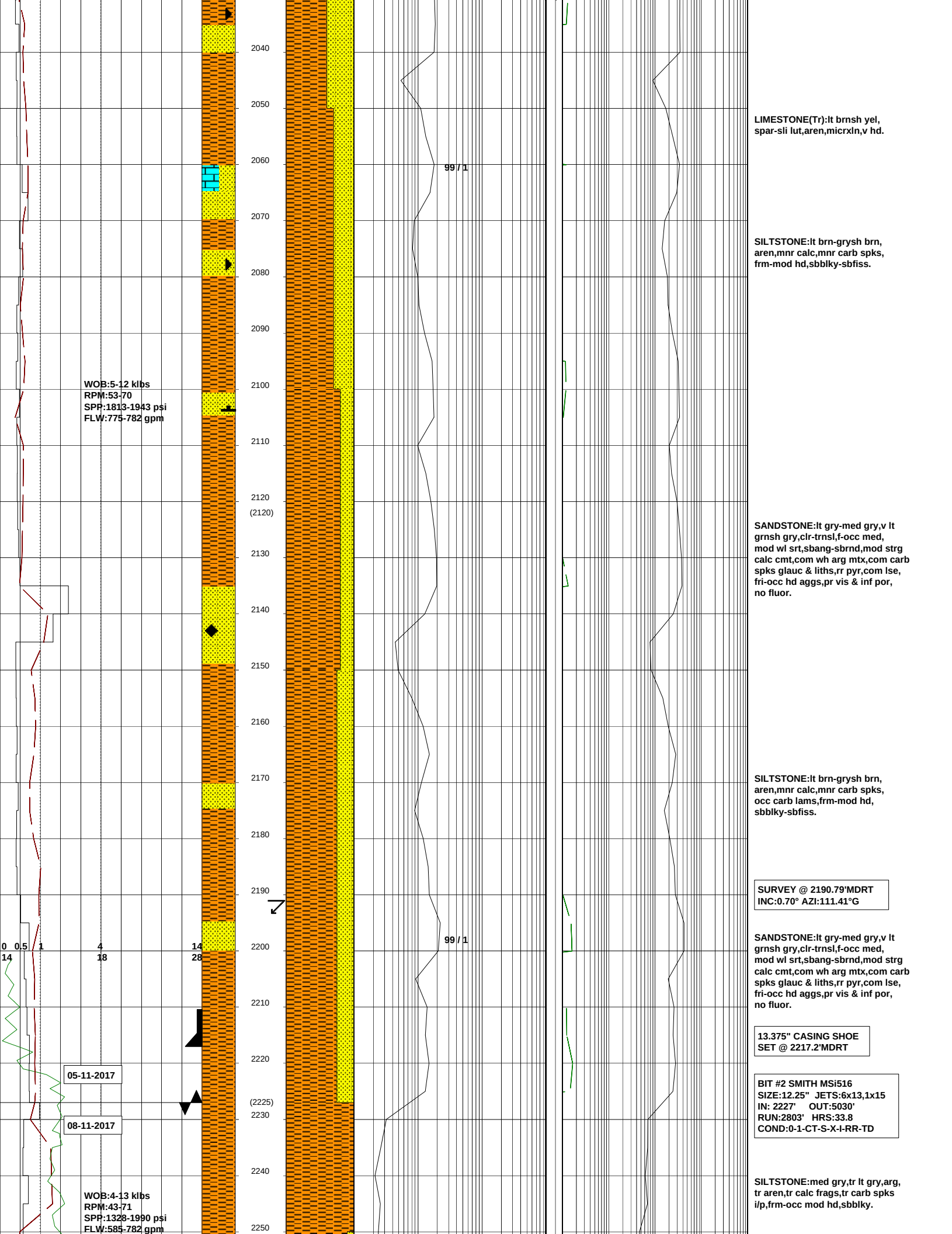
SANDSTONE:lt gry,off wh,mott gry,vf-f,mod wl srt,sbang-sbrnd, mod sil cmt,occ strg calc cmt, com lt gry arg mtx,tr liths,tr glauc, fri aggs,pr-fr vis por,no fluor.

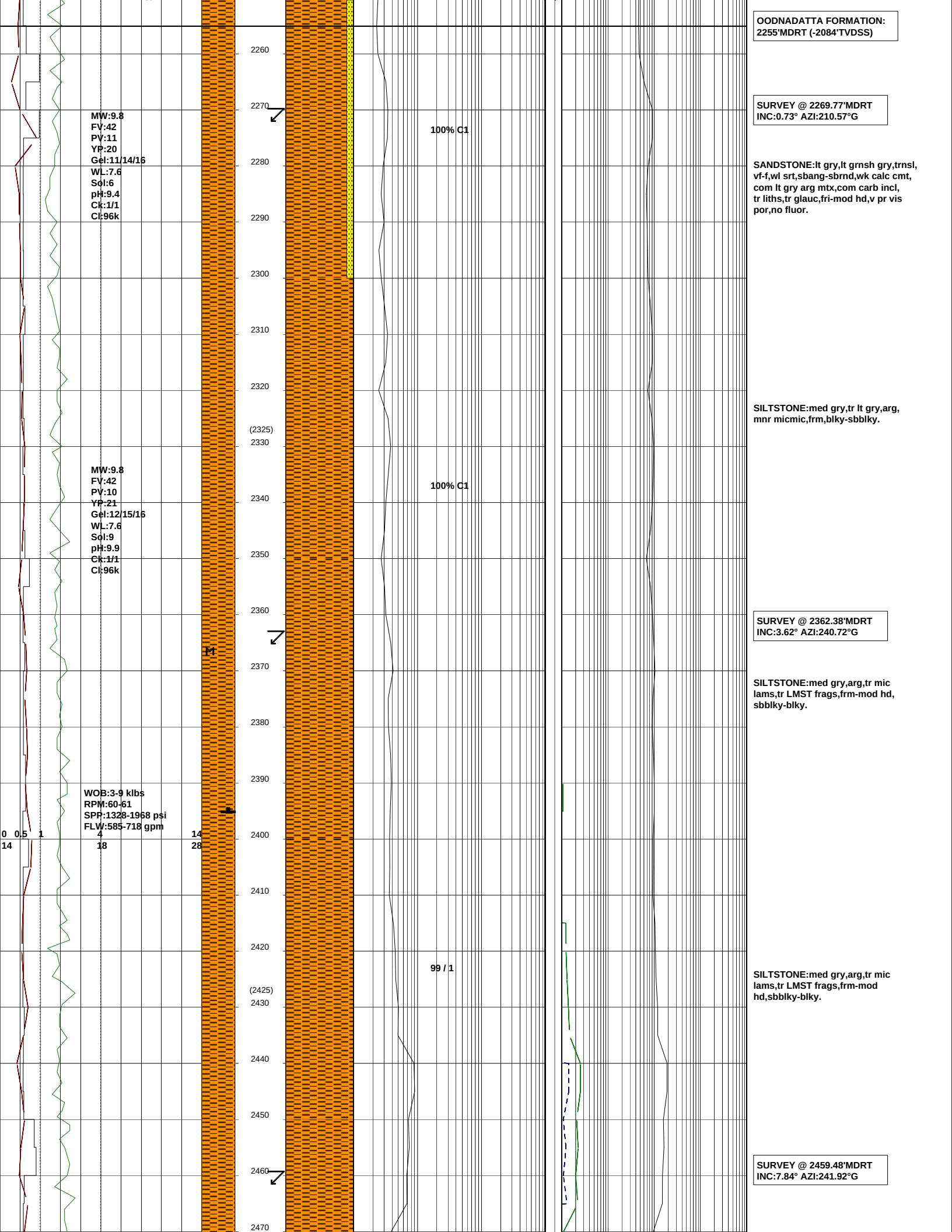


MW:	9.4
FV:	44
PV:	11
YP:	23
Gen:	12/15/17
WL:	7.6
Sol:	5
pH:	9.6
Ck:	1
Cl:	49k



SILTSTONE:lt brn-grysh brn,
aren,mnr calc,mnr carb spks,
frm-mod hd.sbbllky-sbflss.





OODNADATTA FORMATION:
2255'MDRT (-2084'TVDSS)

SURVEY @ 2269.77'MDRT
INC:0.73° AZI:210.57°G

SANDSTONE:lt gry,lt grnsh gry,trnsl,
vf-f,wl srt,sbang-sbrnd,wk calc cmt,
com lt gry arg mtx,com carb incl,
tr liths,tr glauc,fri-mod hd,v pr vis
por,no fluor.

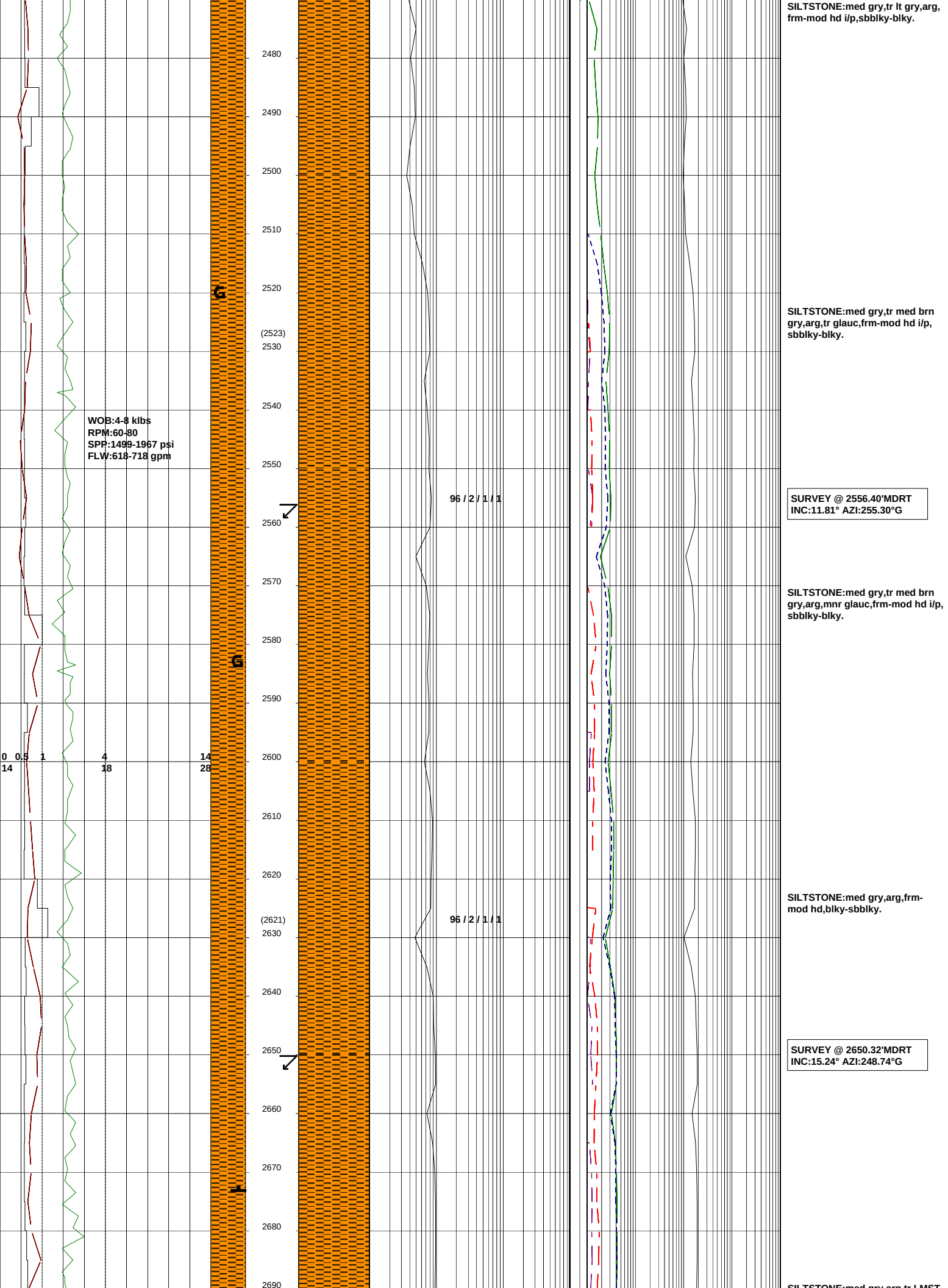
SILTSTONE:med gry,tr lt gry,arg,
mnr micmic,frm,blky-sbblky.

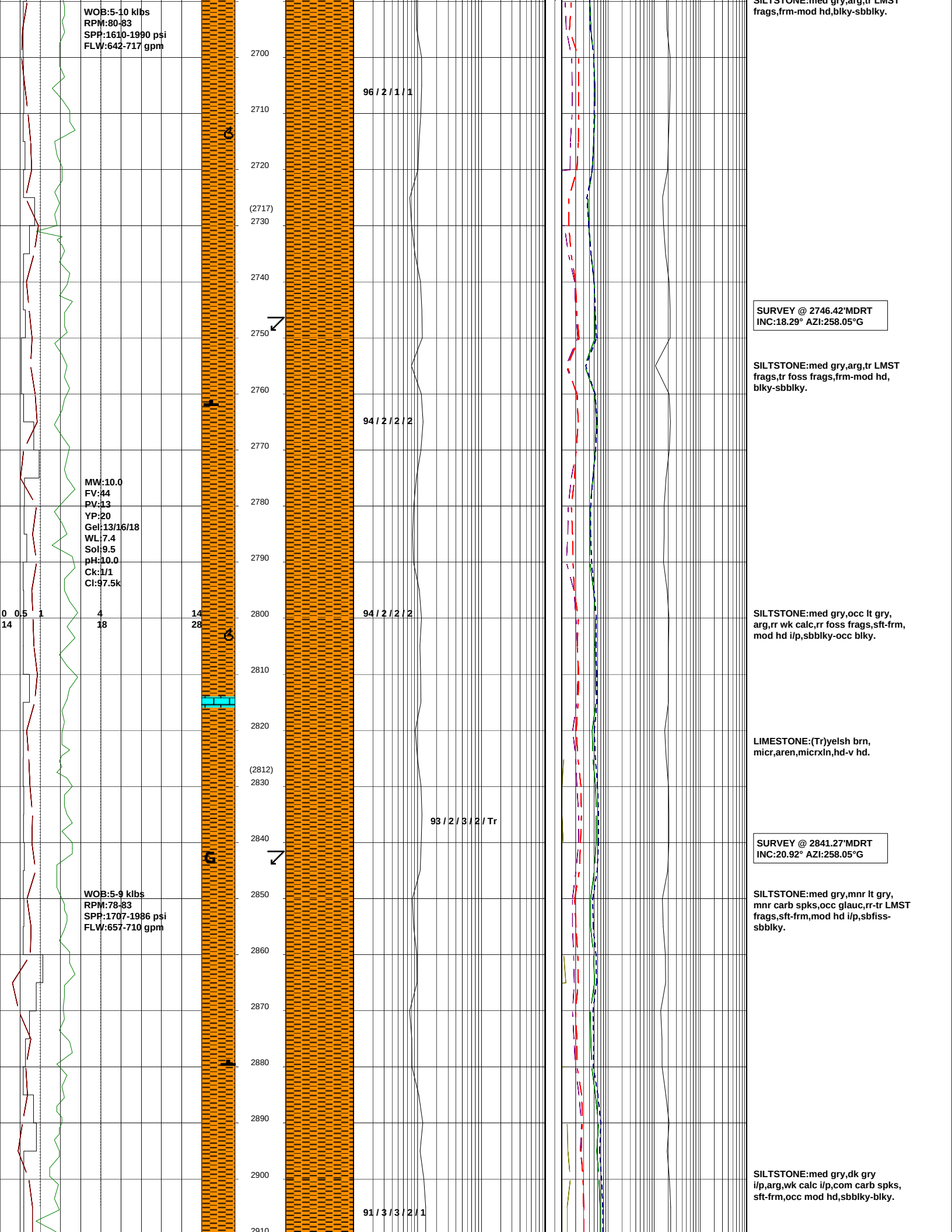
SURVEY @ 2362.38'MDRT
INC:3.62° AZI:240.72°G

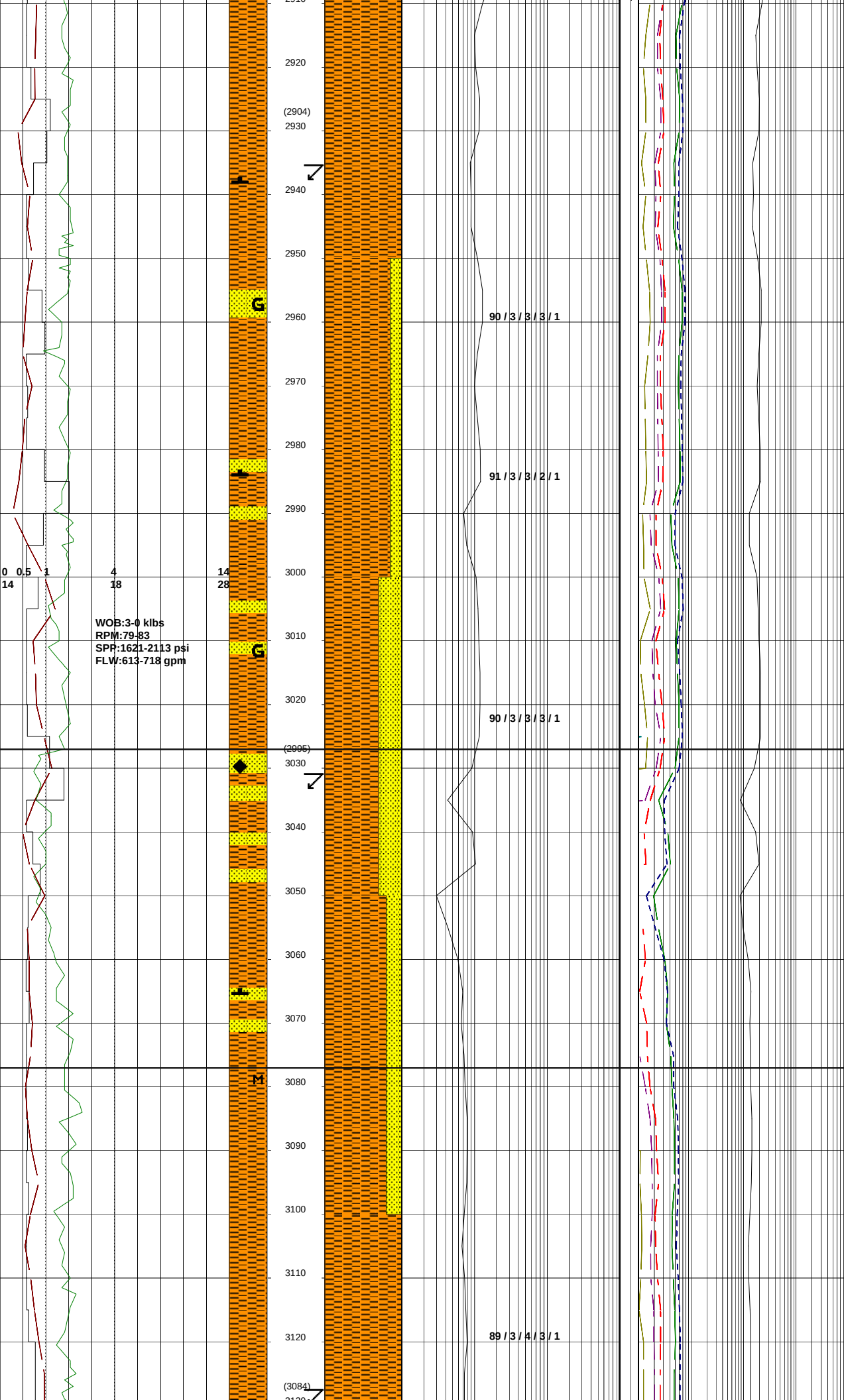
SILTSTONE:med gry,arg,tr mic
lams,tr LMST frags,frm-mod hd,
sbblky-blky.

SILTSTONE:med gry,arg,tr mic
lams,tr LMST frags,frm-mod
hd,sbblky-blky.

SURVEY @ 2459.48'MDRT
INC:7.84° AZI:241.92°G







SURVEY @ 2935.83'MDRT
INC:23.81° AZI:260.27°G

SILTSTONE: med gry, lt gry,
arg, mnr aren, com carb spks, occ
carb micrlams, sft frm, occ mod hd,
blkly-sbblky.

90 / 3 / 3 / 3 / 1

91 / 3 / 3 / 2 / 1

SANDSTONE: lt-med gry, tr off
wh, f-vf, wl srt, sbang-sbrnd, wk calc
cmt, com lt-med gry slty mt, tr off
wh arg mt, occ carb spks & flks,
mnr glauc grs, tr nod pyr, fri aggs,
pr-ti vis por, no fluor.

WOB: 3-0 klbs
RPM: 79-83
SPP: 1621-2113 psi
FLW: 613-718 gpm

90 / 3 / 3 / 3 / 1

COORIKIANA SANDSTONE:
3027'MDRT (-2826'TVDSS)

SURVEY @ 3030.84'MDRT
INC:26.53° AZI:263.70°G

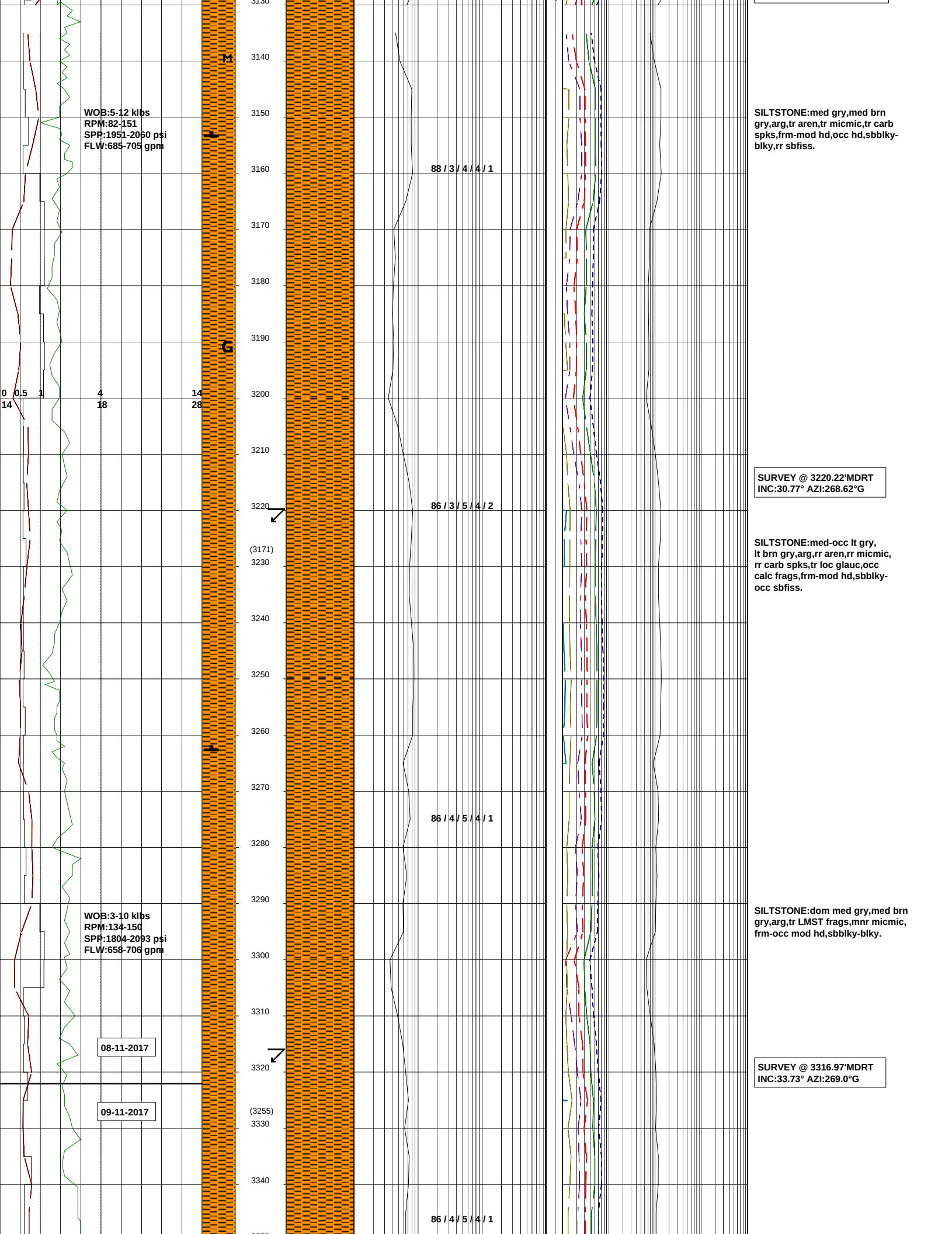
SILTSTONE: med-lt gry, arg,
rr carb spks, tr dissem pyr, frm-mod
hd, occ hd, sbblky-blky, rr sbfiss.

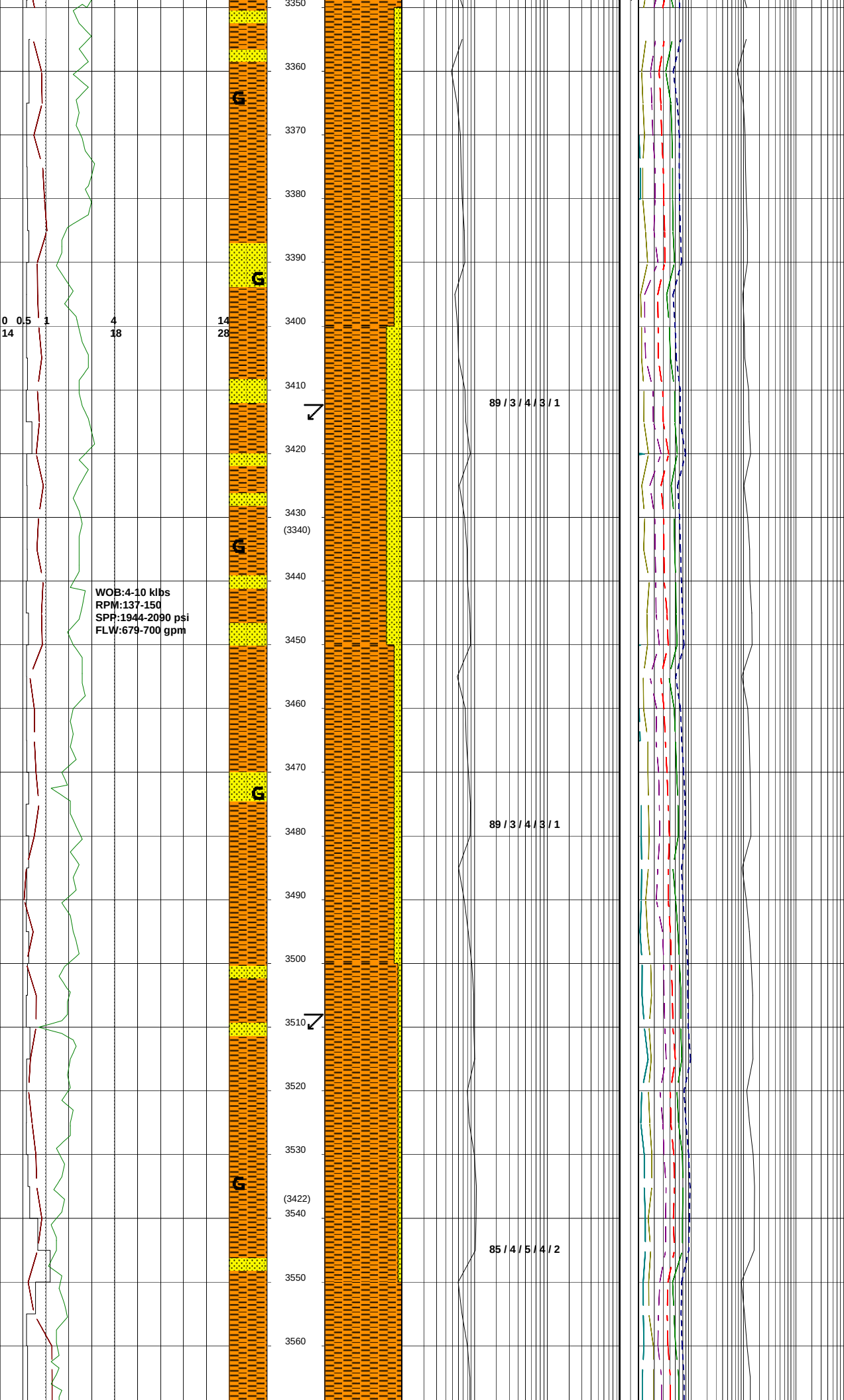
BULLDOG SHALE:
3077'MDRT (-2870'TVDSS)

SANDSTONE: lt-med gry, tr off
wh, f-vf, wl srt, sbang-sbrnd, wk calc
cmt, com lt-med gry slty mt, tr off
wh arg mt, occ carb spks & flks,
mnr glauc grs, tr nod pyr, fri aggs,
pr-ti vis por, no fluor.

89 / 3 / 4 / 3 / 1

SURVEY @ 3127.18'MDRT
INC:28.34° AZI:266.80°G





WOB:4-10 klbs
RPM:137-150
SPR:1944-2090 psi
FLW:679-700 gpm

SANDSTONE:lt-med gry,off wh,vf
grd to aren SLTST,wl srt,sbang-
sbrnd,mod sil cmt,com med gry arg
mtx,com glauc,fri-mod hd,v pr vis
por,no fluor.

SURVEY @ 3412.34'MDRT
INC:36.83° AZI:268.55°G

SILTSTONE:dom med gry,med brn
gry,arg,tr LMST frags,mnr micmic,
loc com glauc,frm-occ mod hd,
sbbiky-blky.

SANDSTONE:med gry,lt grnsh gry,
tr off wh,vf,grd to aren SLTST,
wl srt,sbang-sbrnd,mod sil cmt,
com med gry arg mtx,com glauc,
fri-occ mod hd,v pr vis por,no
fluor.

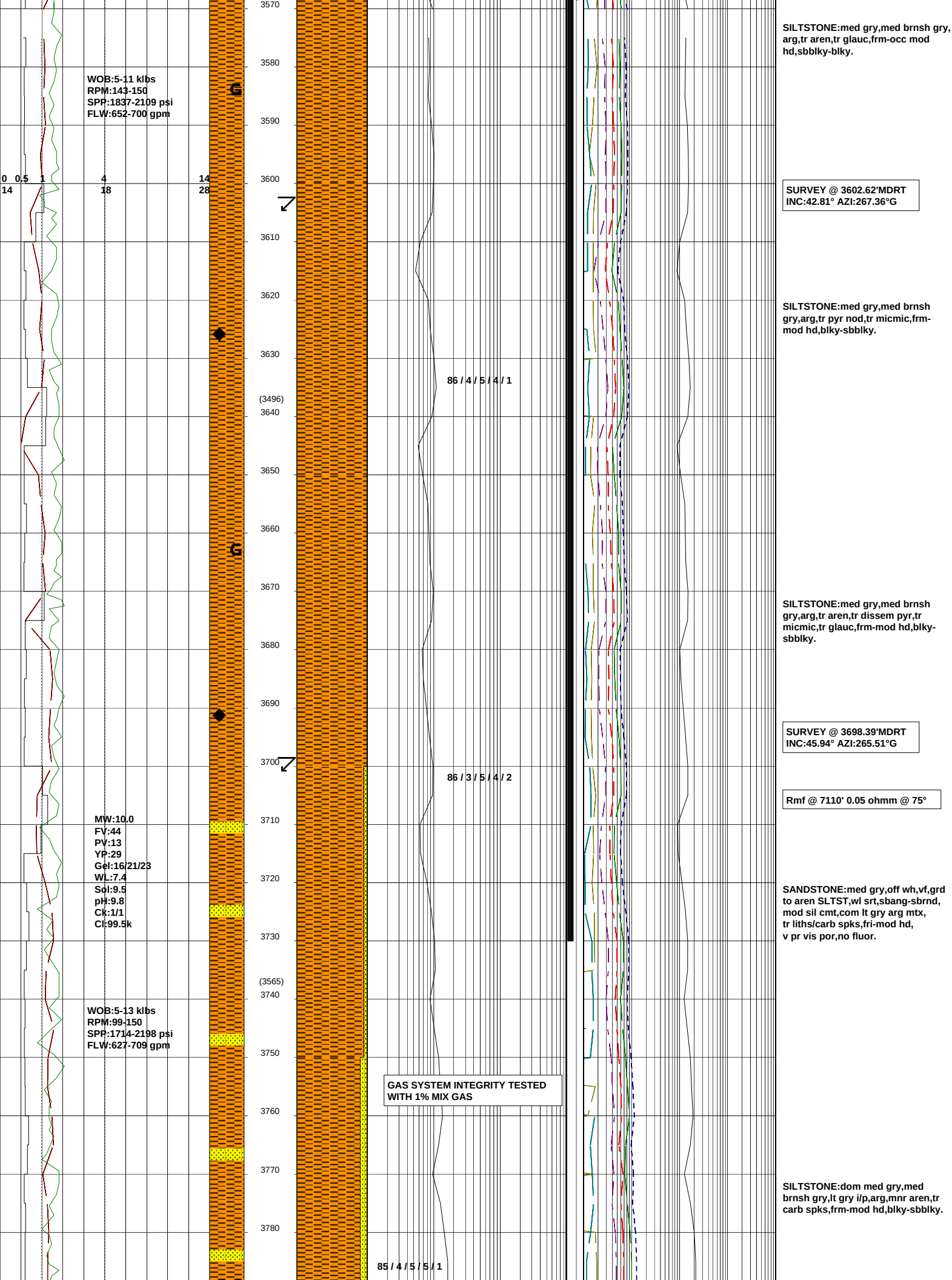
SURVEY @ 3507.9'MDRT
INC:39.89° AZI:267.98°G

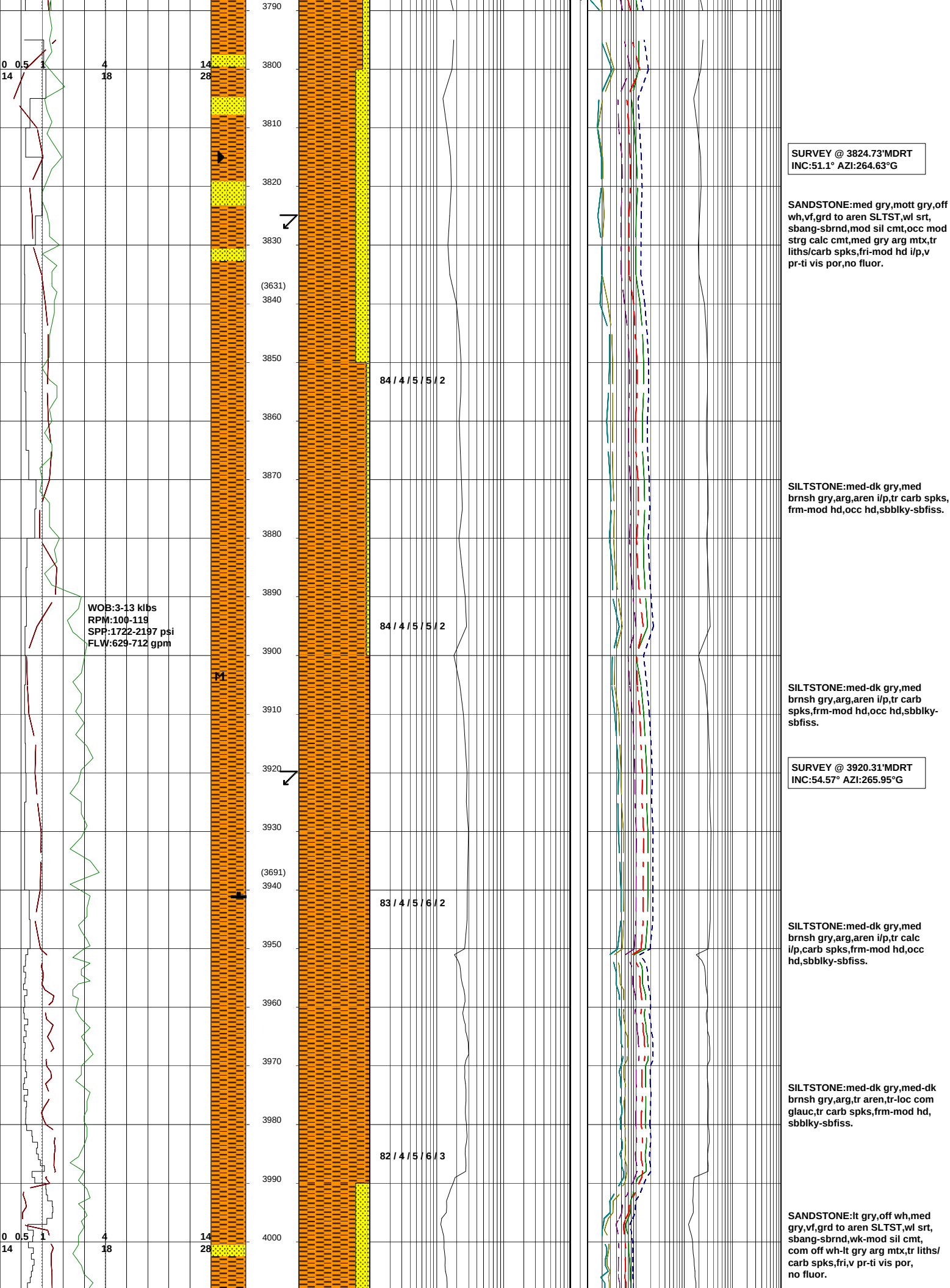
SILTSTONE:med gry,med brnsh gry,
arg,tr-loc com glauc,mnr micmic,
frm-tr mod hd,sbbiky-blky.

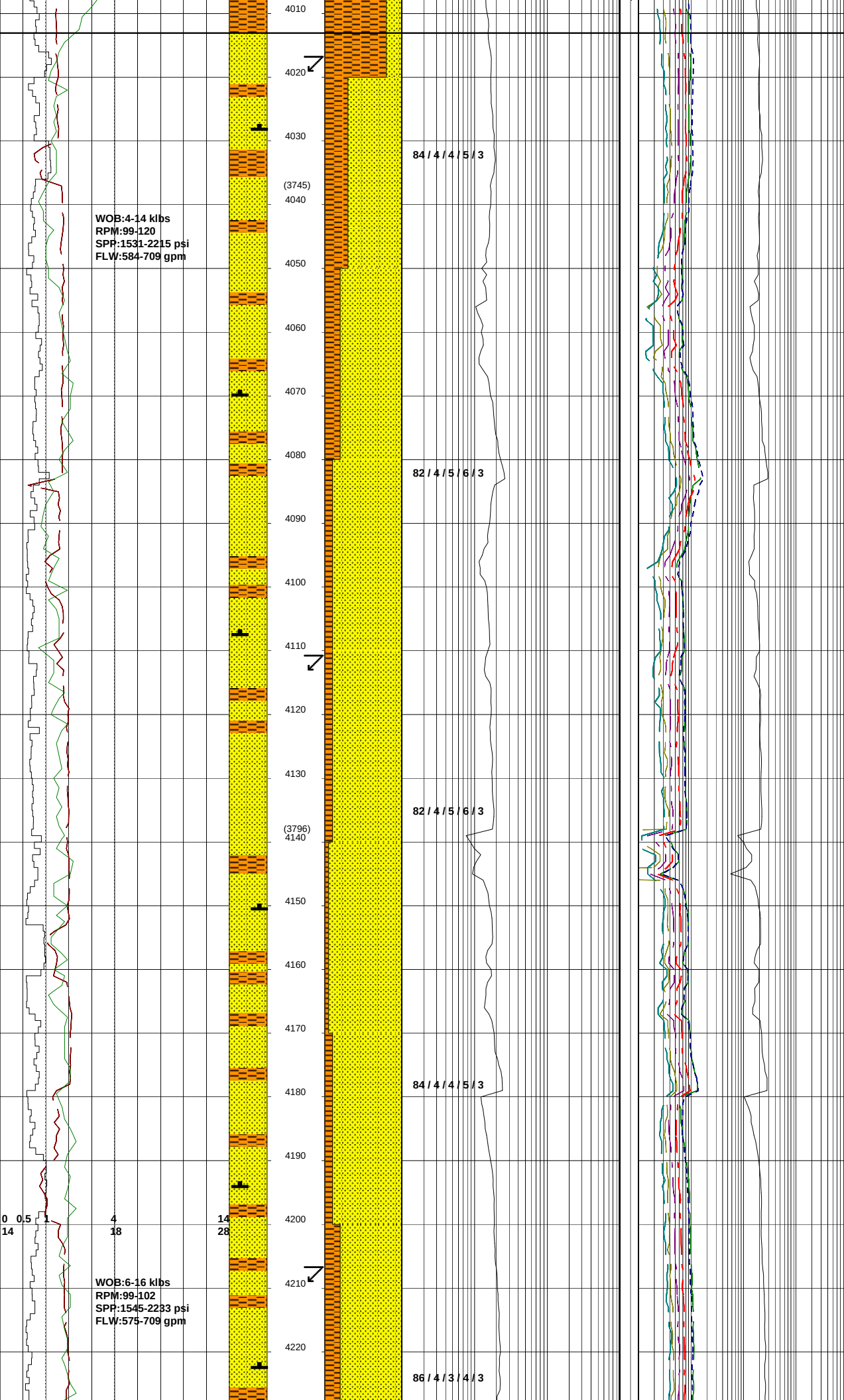
89 / 3 / 4 / 3 / 1

89 / 3 / 4 / 3 / 1

85 / 4 / 5 / 4 / 2







WOB:4-14 klbs
RPM:99-120
SPP:1531-2215 psi
FLW:584-709 gpm

WOB:6-16 klbs
RPM:99-102
SPP:1545-2233 psi
FLW:575-709 gpm

CADNA-OWIE FORMATION:
4013'MDRT (-3563'TVDSS)

SURVEY @ 4016.36'MDRT
INC:57.58° AZI:267.25°G

SANDSTONE:off wh,lt gry,rr
lt brn gry,f-vf,wl srt,sbang-sbrnd,
wk-occ mod strg calc cmt,mod abdt
off wh arg mtx,mnr-rr lt gry-lt brn
gry slty mtx,occ carb spks,occ liths,
fri-occ mod strg aggs,pr-ti vis por,
no fluor.

SILTSTONE:lt-med gry,lt-med
brnsh gry,lt-med brn,aren,occ arg,
mnr carb spks,mod hd,blky-sbbkly.

SURVEY @ 4110.56'MDRT
INC:60.64° AZI:266.96°G

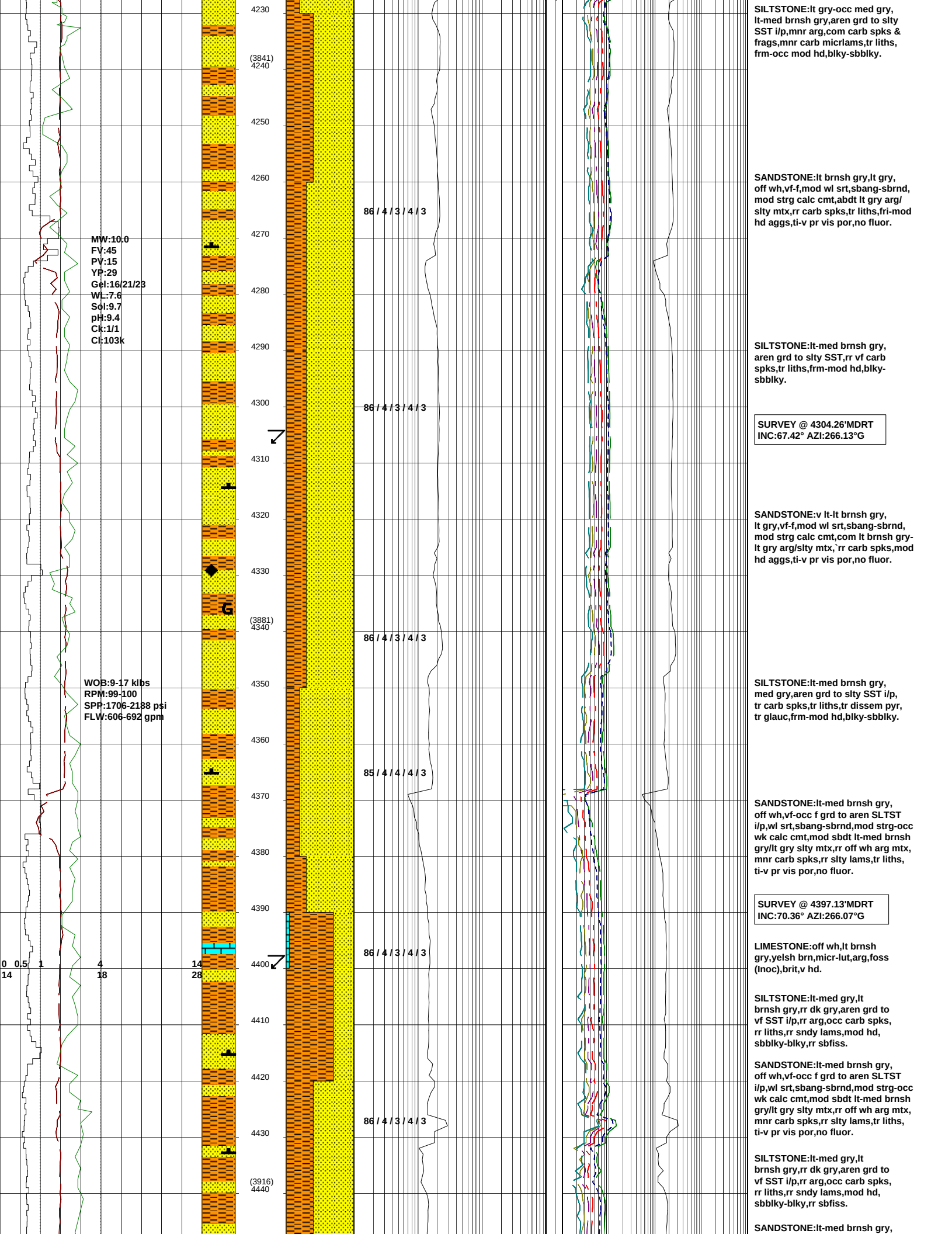
SANDSTONE:v lt gry,v lt brnsh
gry,off wh,vf-f,mod wl srt,sbang-
sbrnd,mod strg calc cmt,com off
wh arg mtx,rr carb spks/frags,tr
liths,fri-mod hd aggs,v pr vis por,
no fluor.

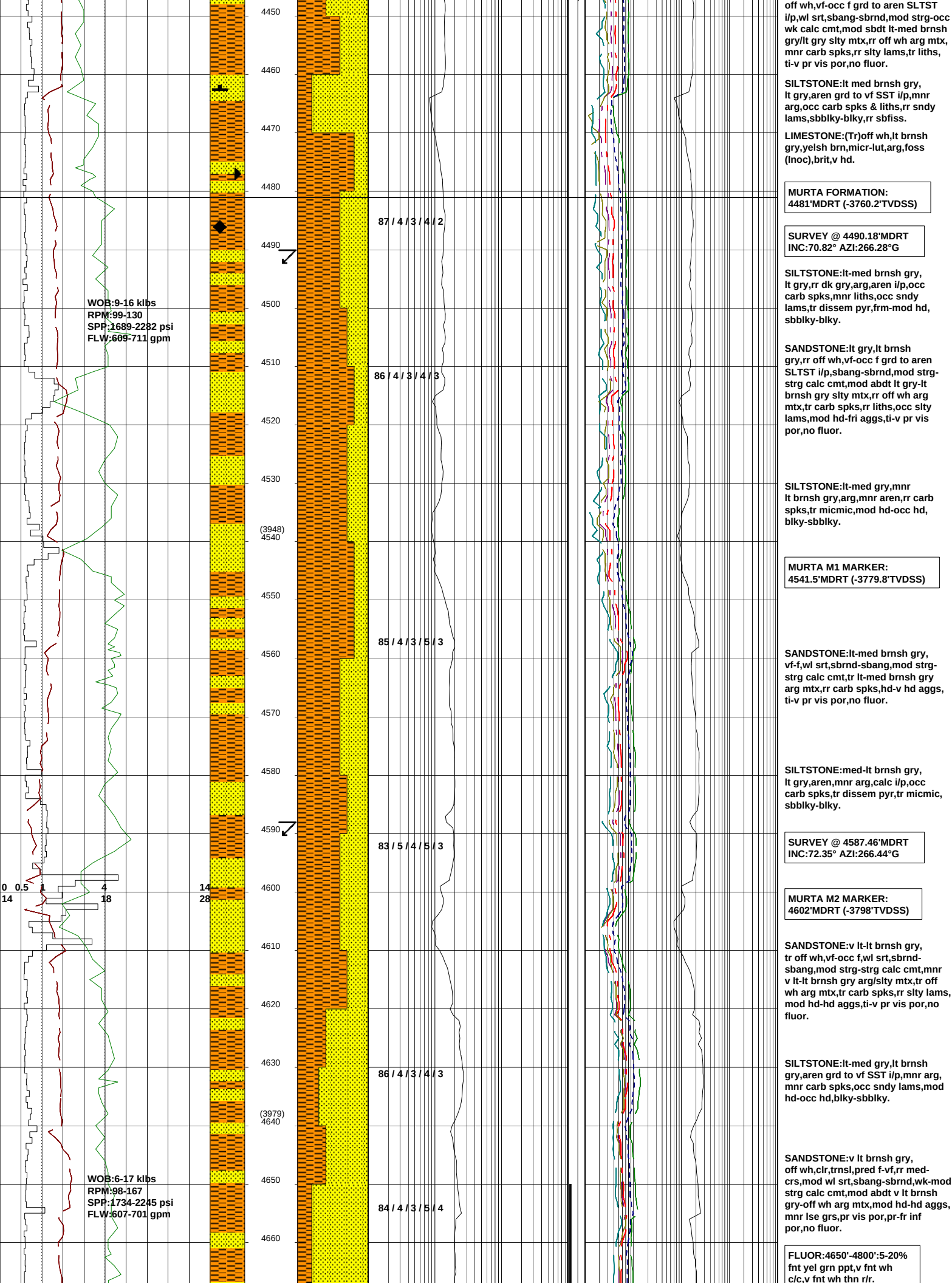
SILTSTONE:lt-med gry,lt-med
brnsh gry,lt-med brn,aren,occ arg,
mnr carb spks,mod hd,blky-sbbkly.

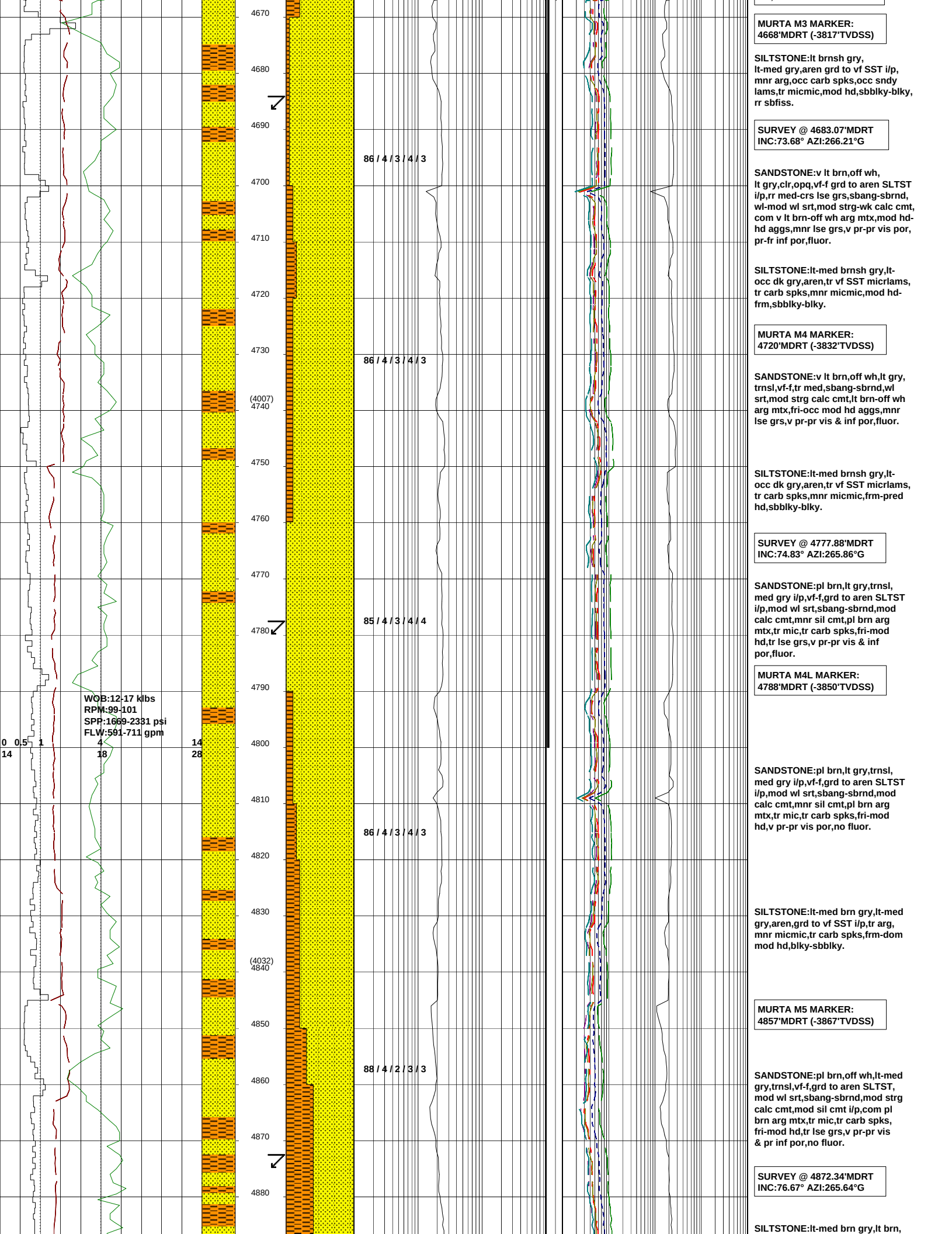
SANDSTONE:off wh,lt gry,lt
brnsh gry,vf-f,wl srt,sbang-sbrnd,
mod strg-occ wk calc cmt,com off
wh arg mtx,mnr lt gry-lt brnsh gry
slty mtx,rr carb spks & frags,tr
liths,mod hd-occ fri aggs,v pr vis
por,no fluor.

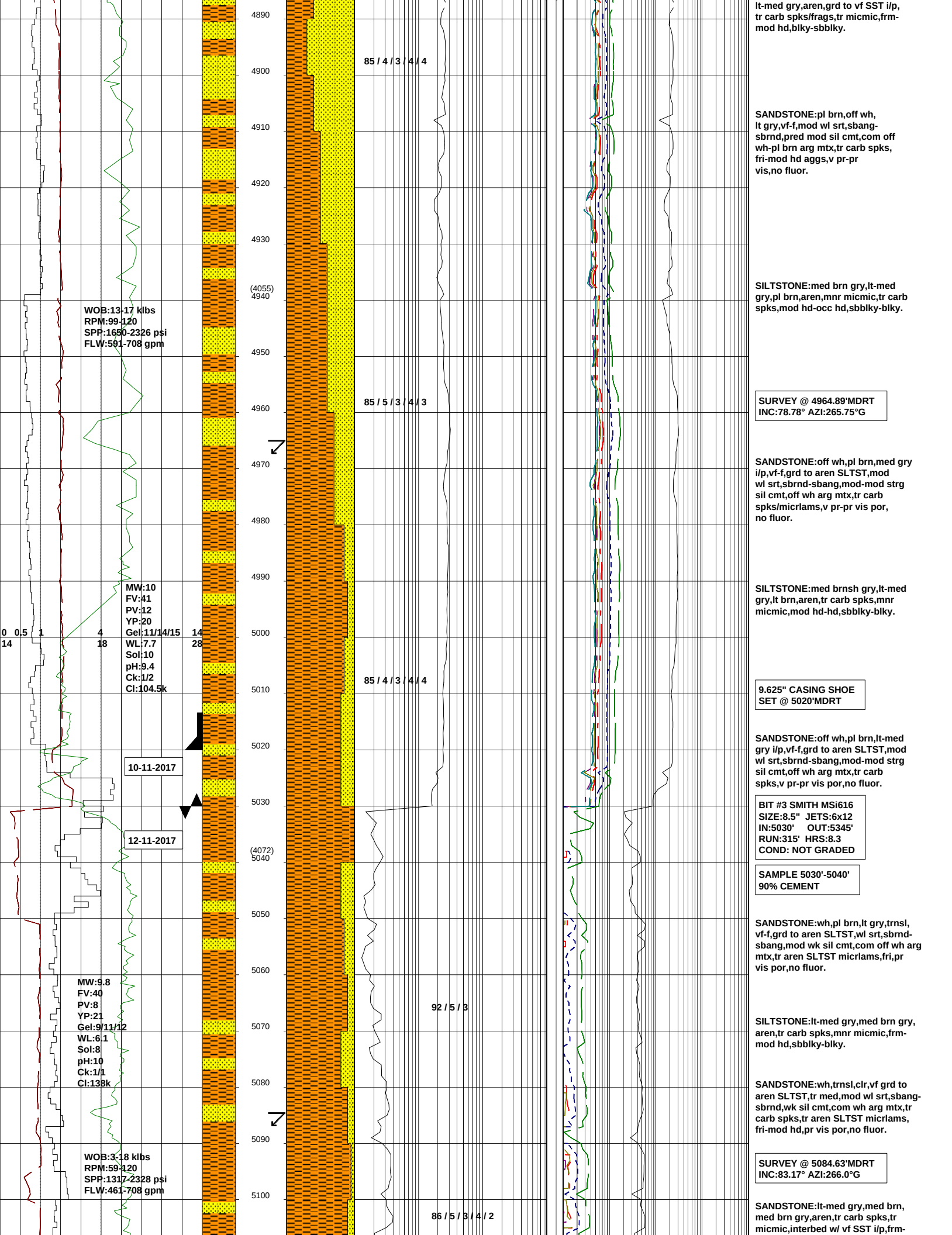
SANDSTONE:lt gry,lt brnsh gry,
mnr off wh,vf-f grd to vf aren SLTST
i/p,wl srt,sbang-sbrnd,mod strg-wk
calc cmt,com lt gry-lt brnsh gry slty
mtx,mnr off wh arg mtx,rr crab spks
& frags,tr liths,mod hd-fri aggs,v
pr vis por,no fluor.

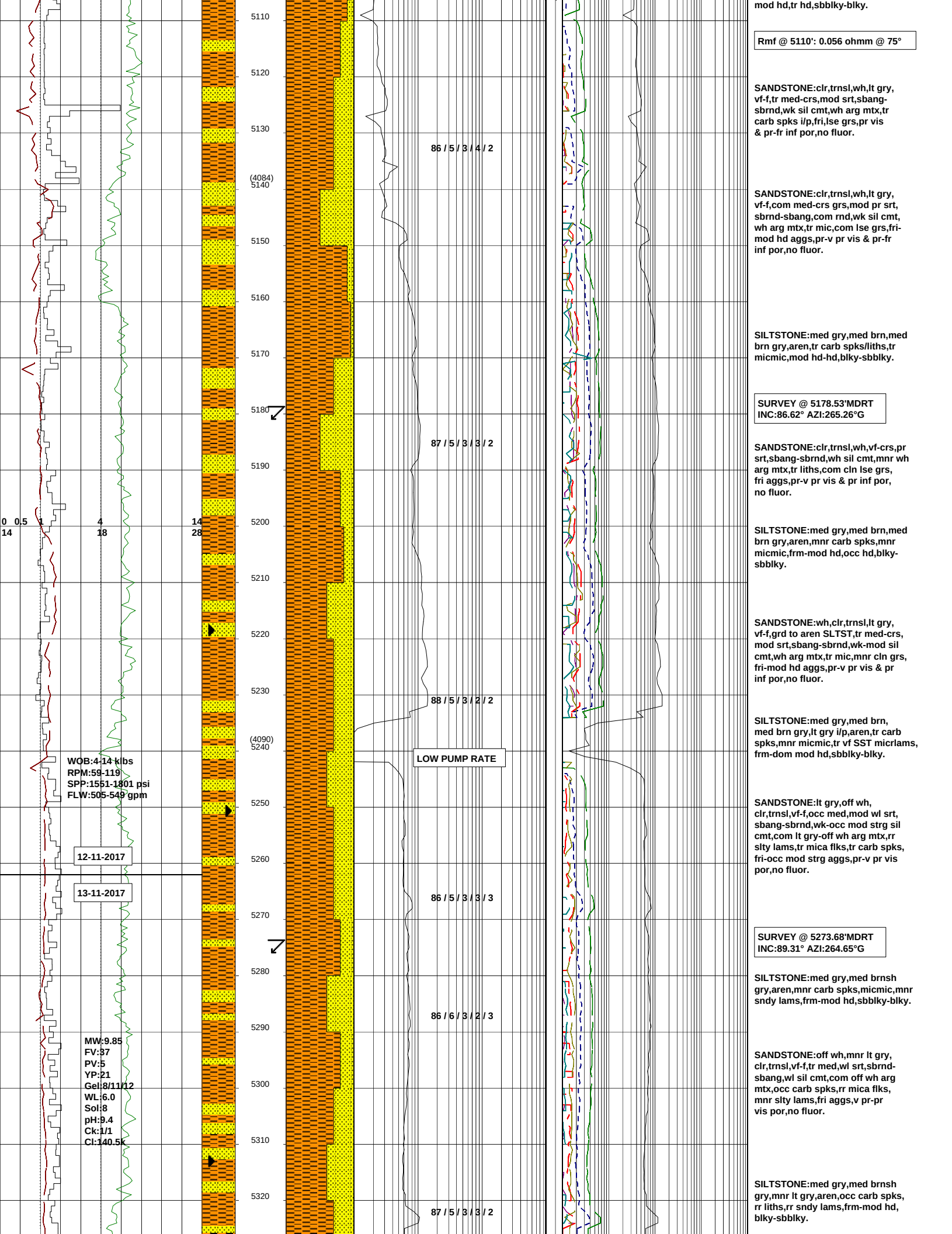
SURVEY @ 4206.89'MDRT
INC:63.88° AZI:266.04°G

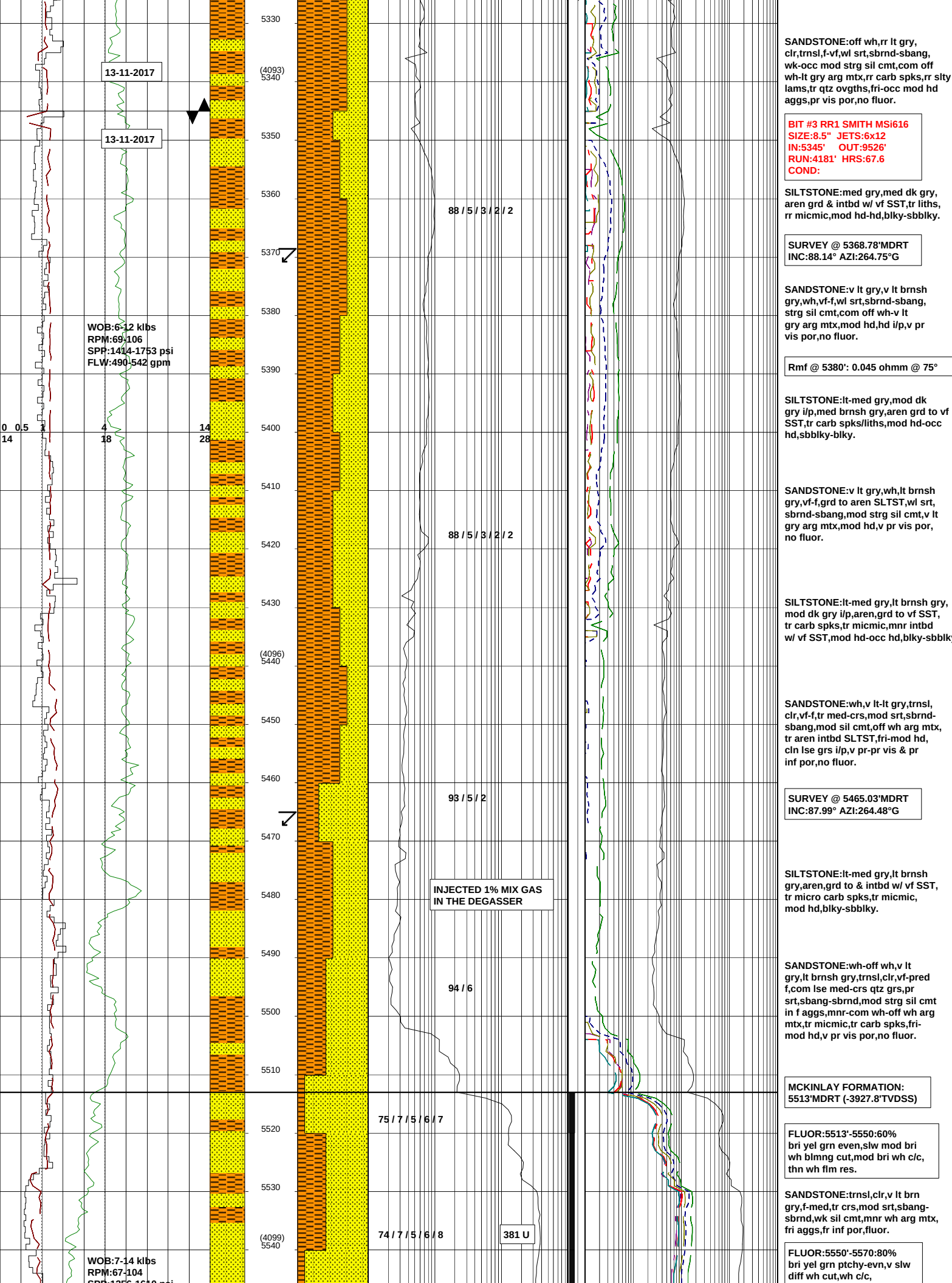


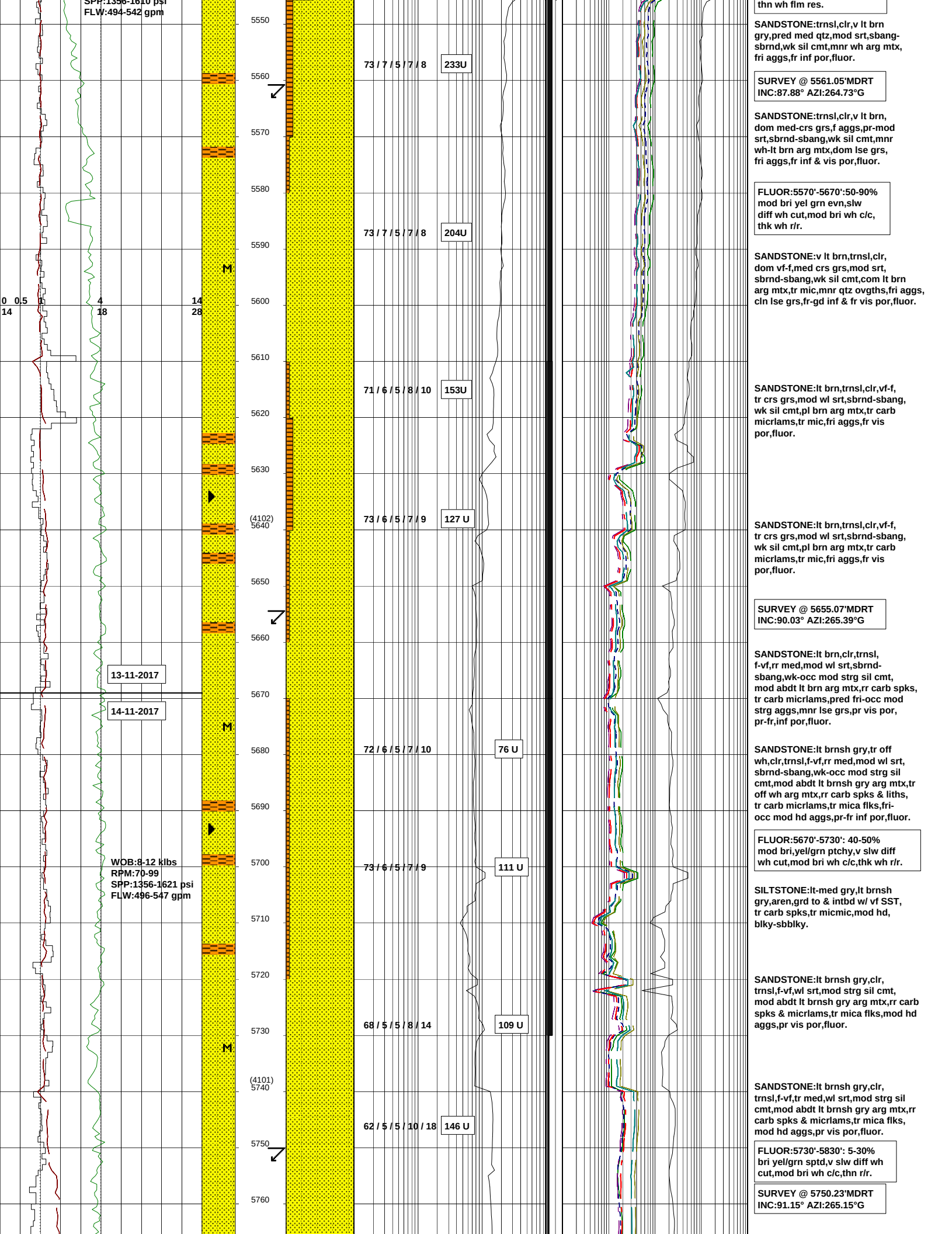


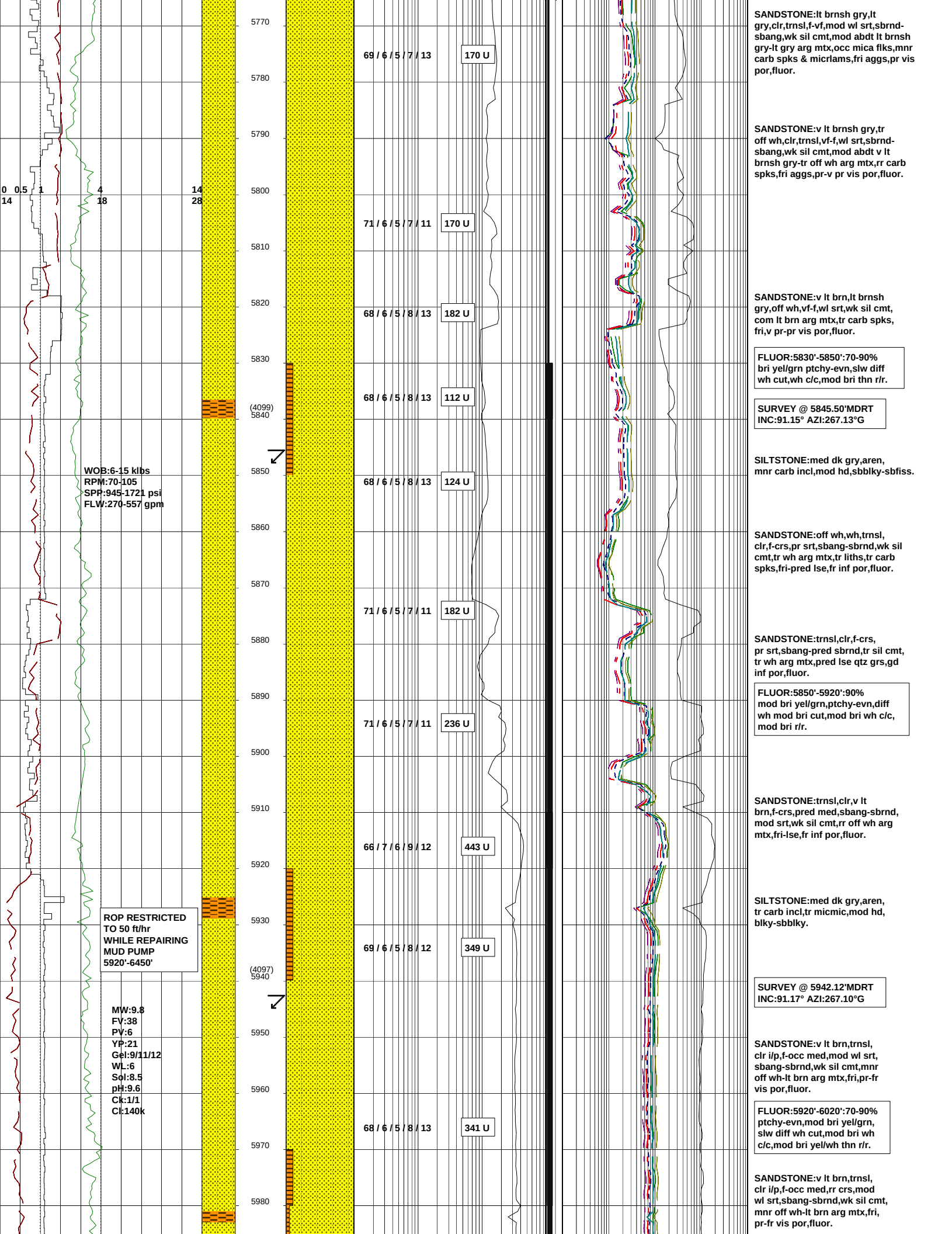


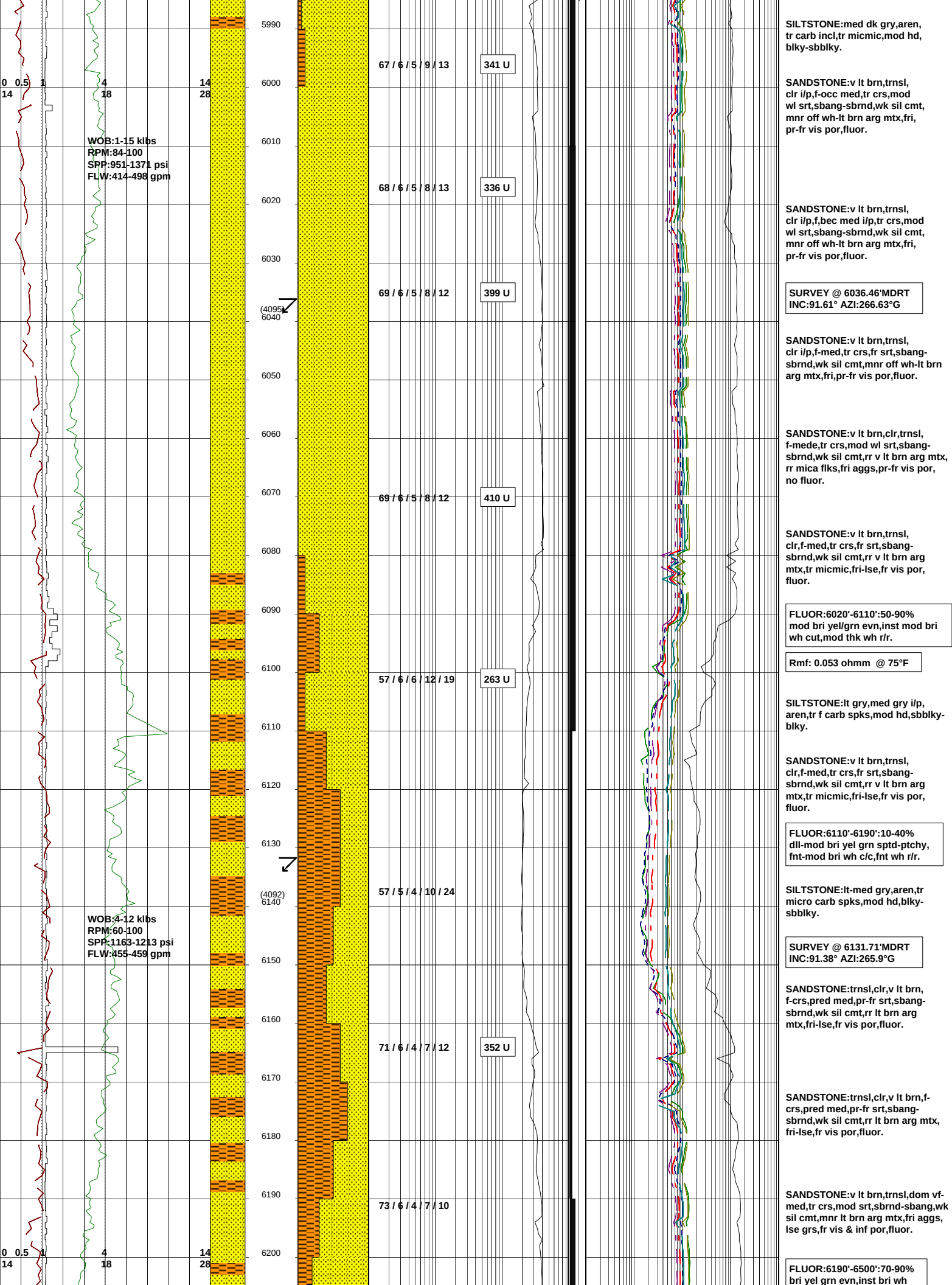


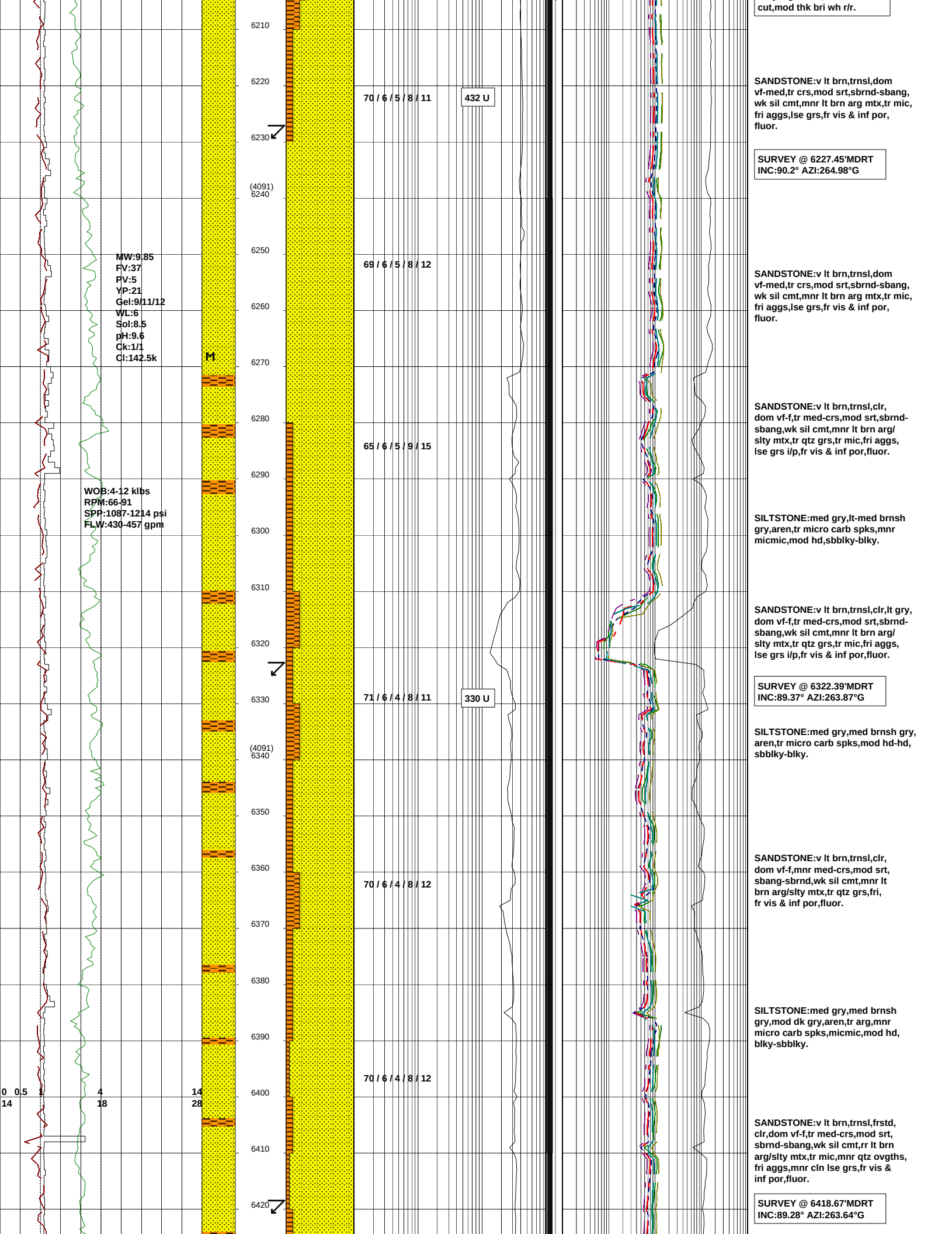


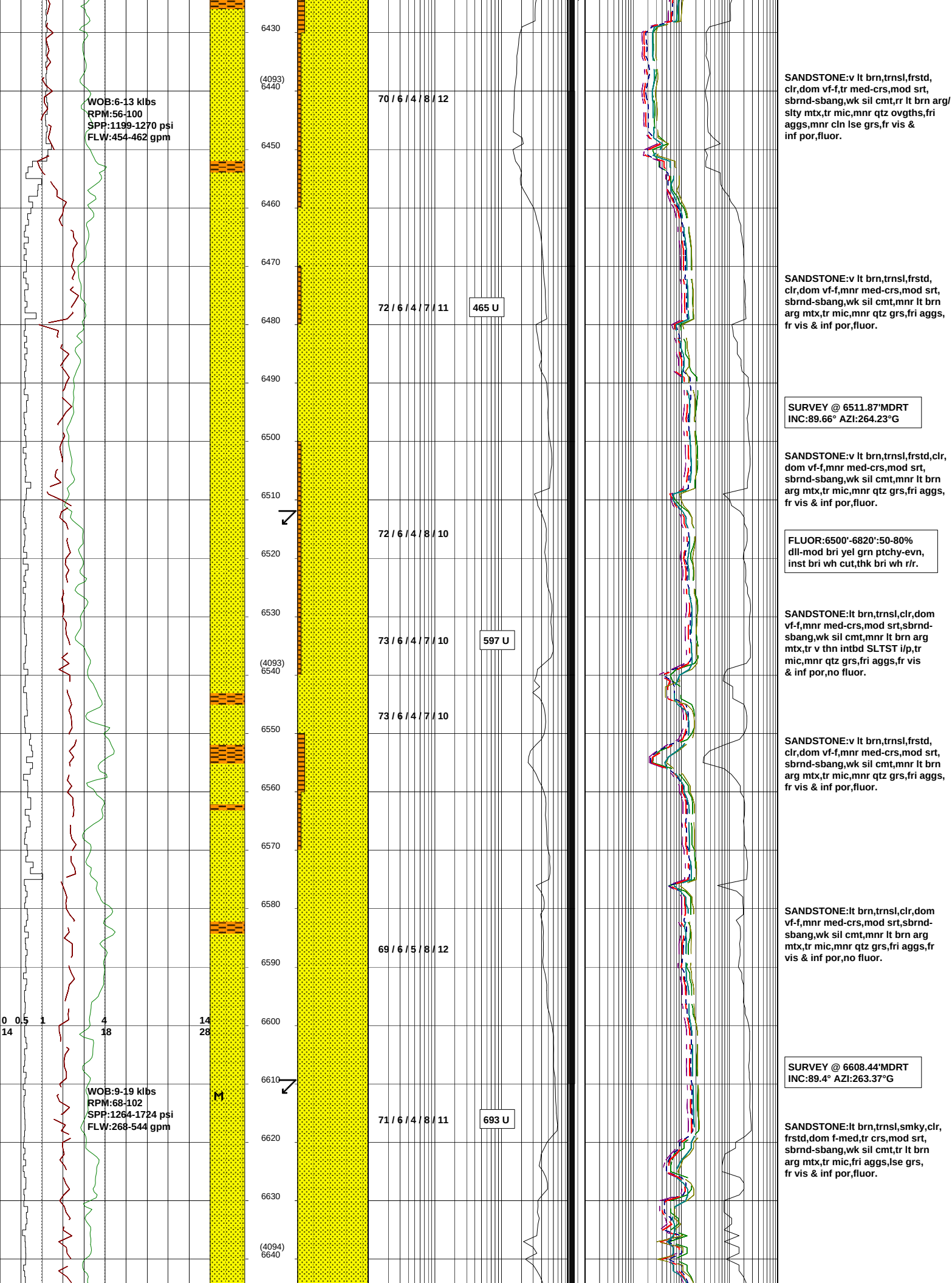


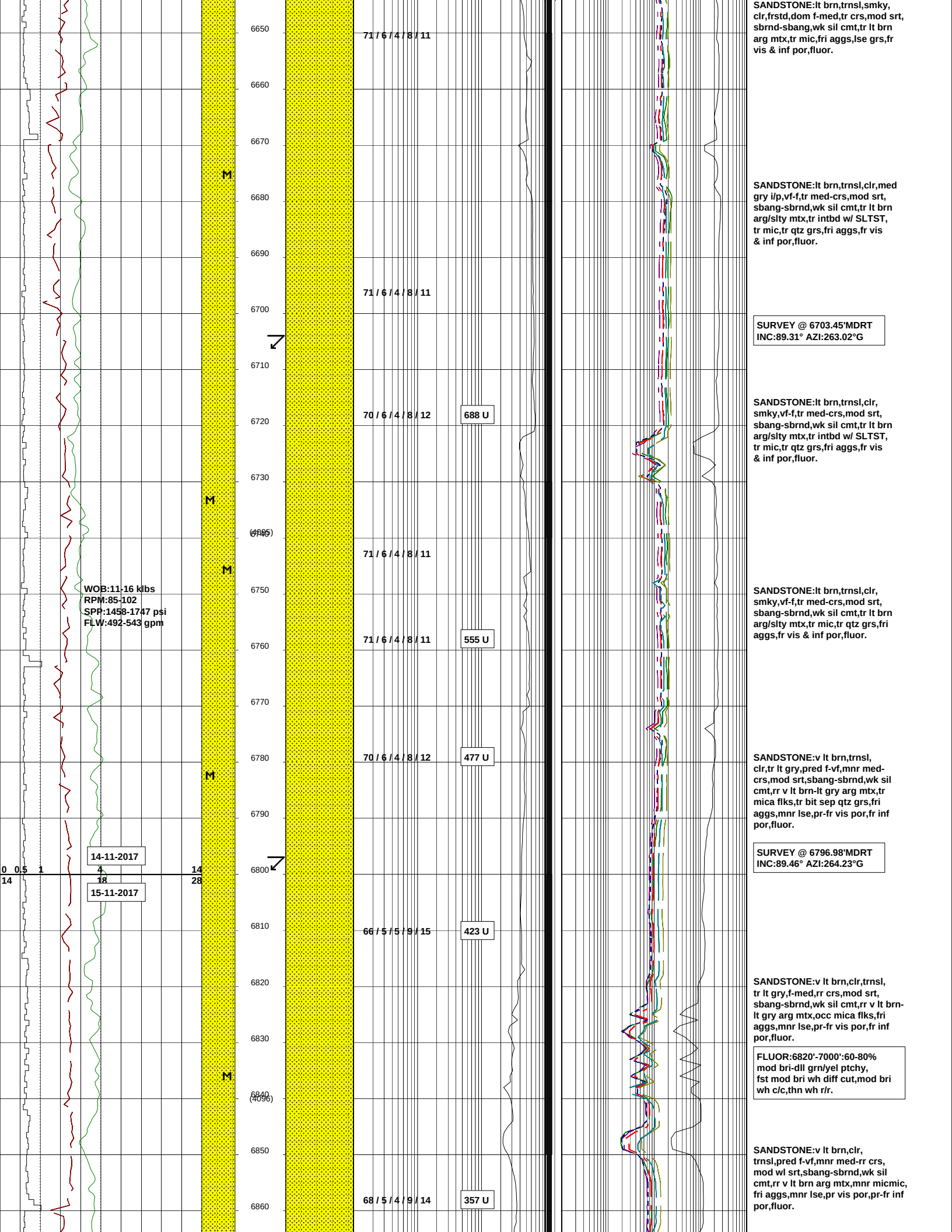


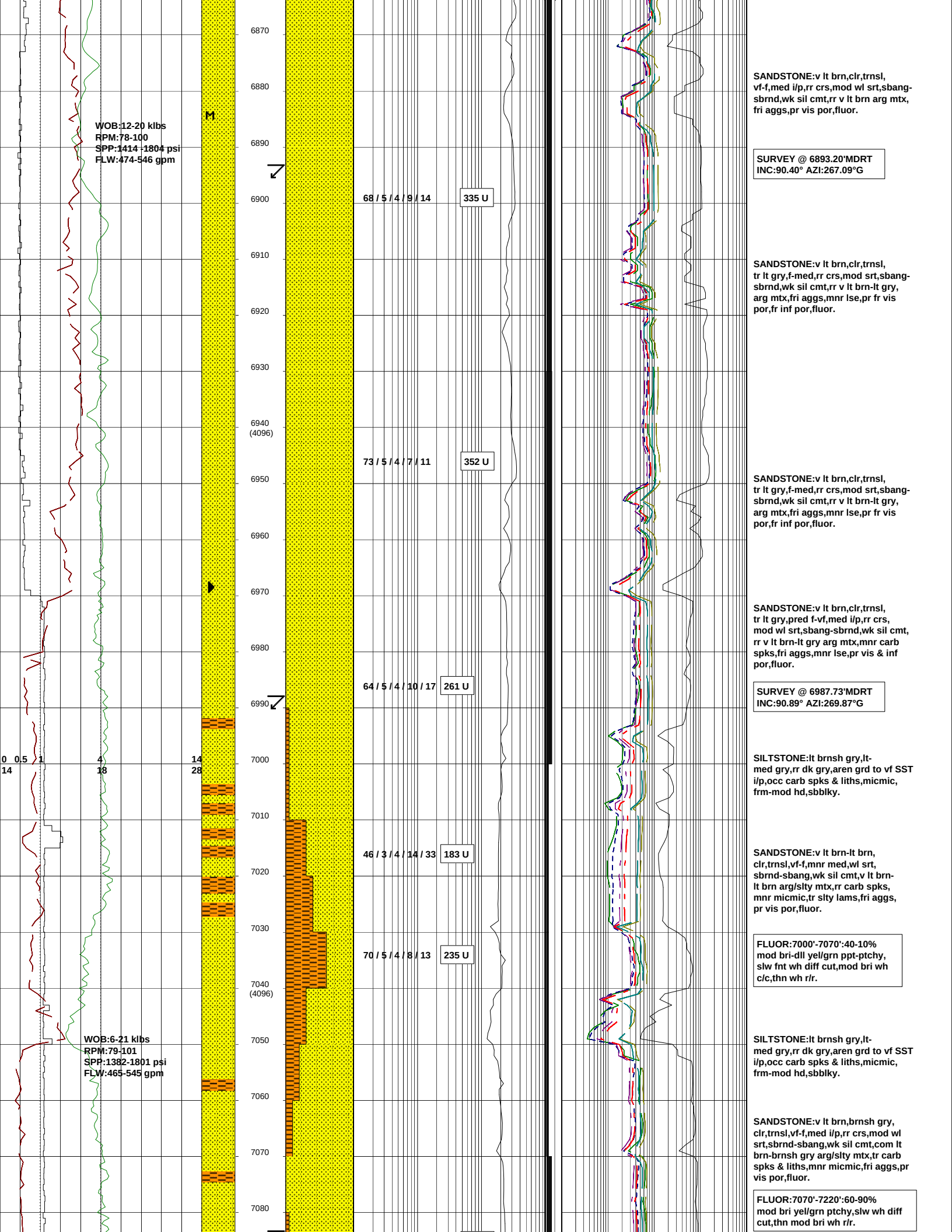


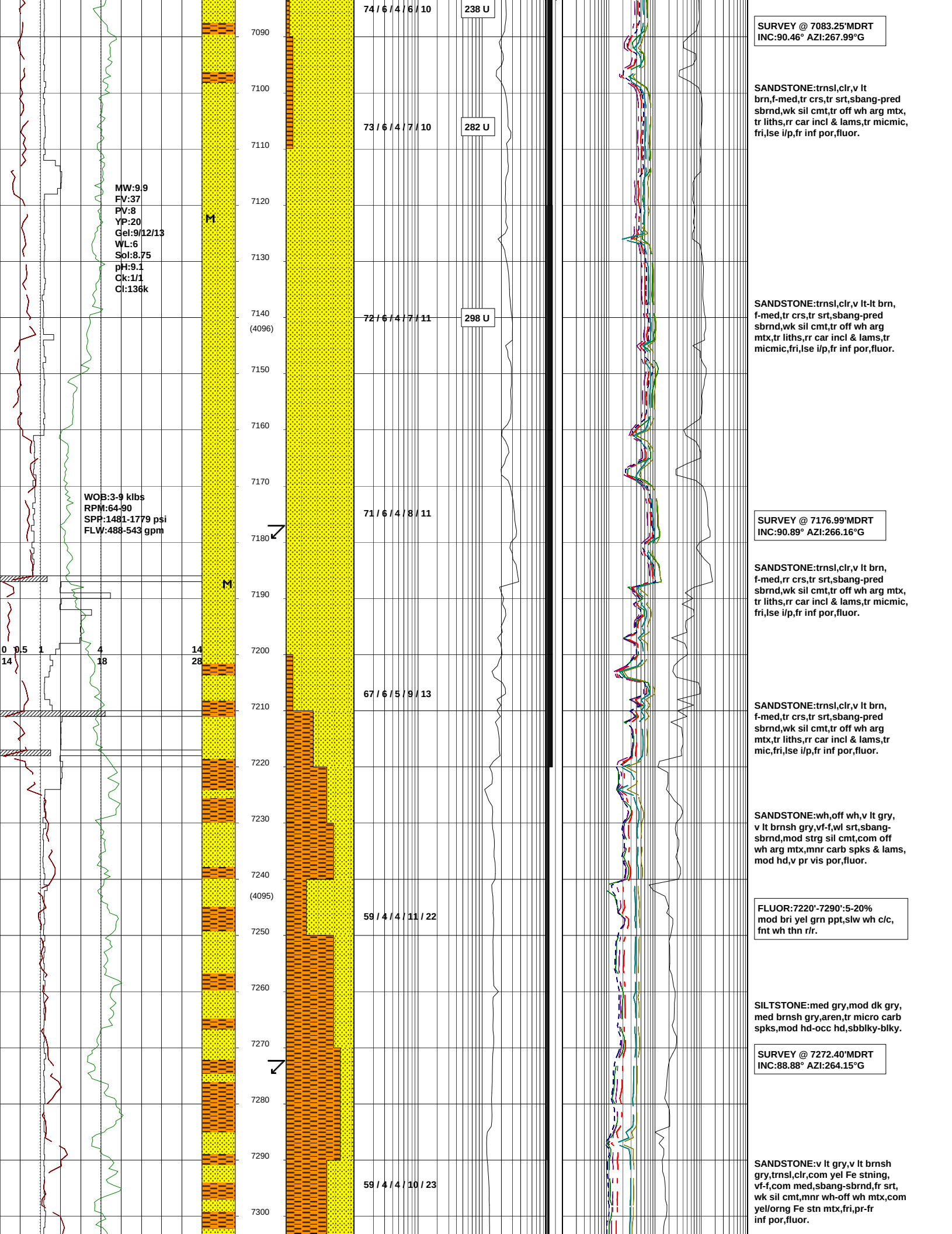


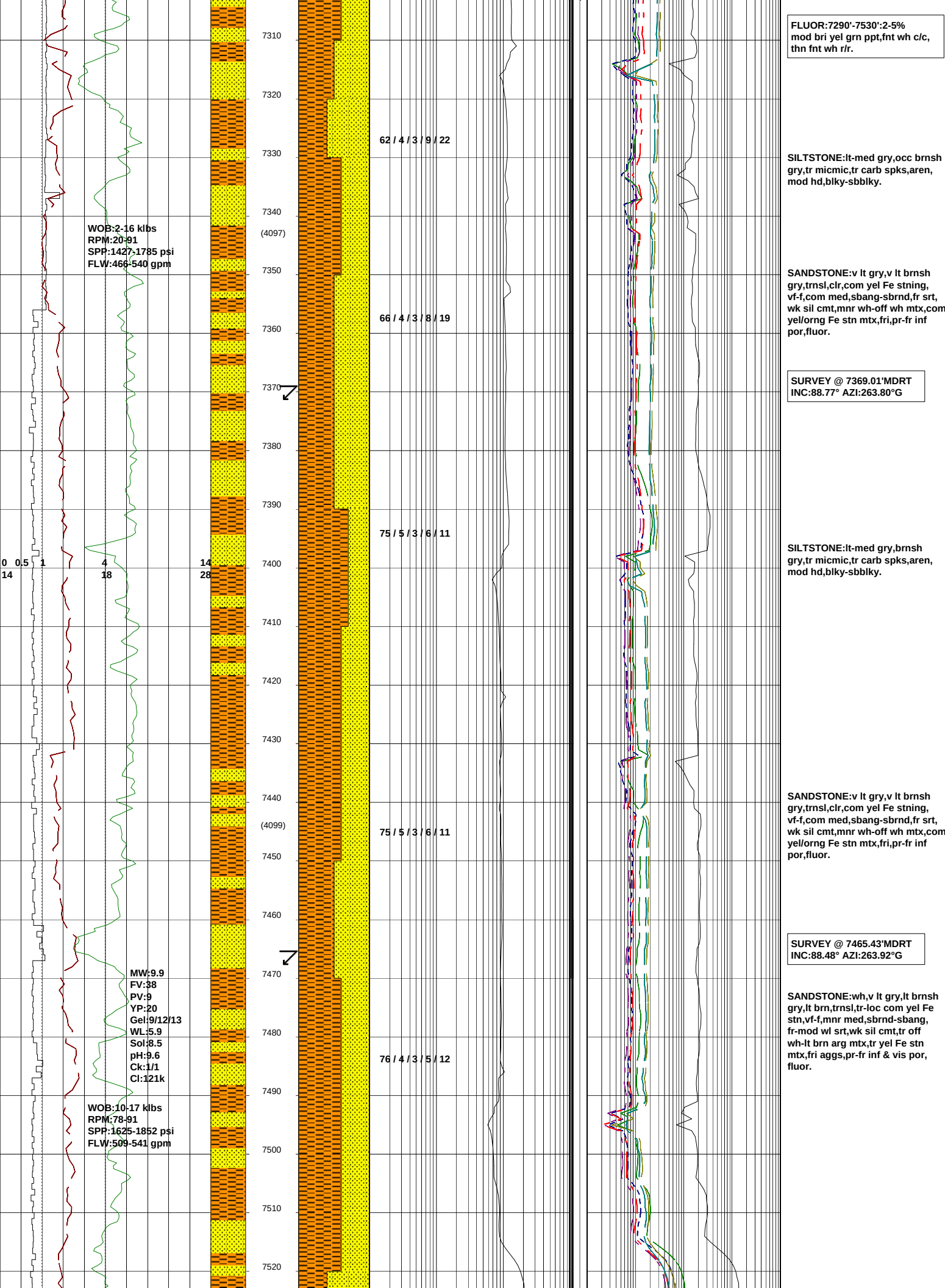


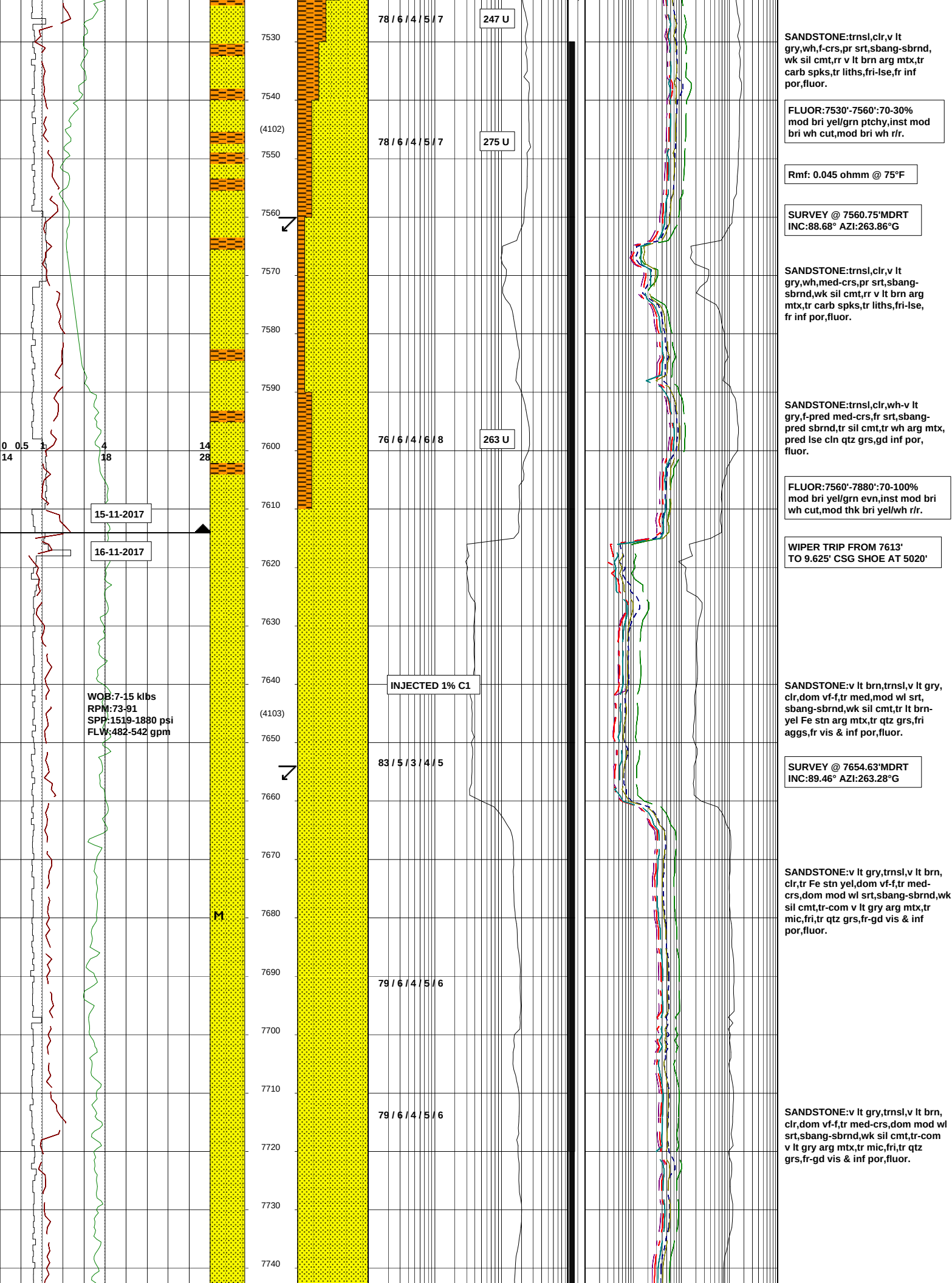


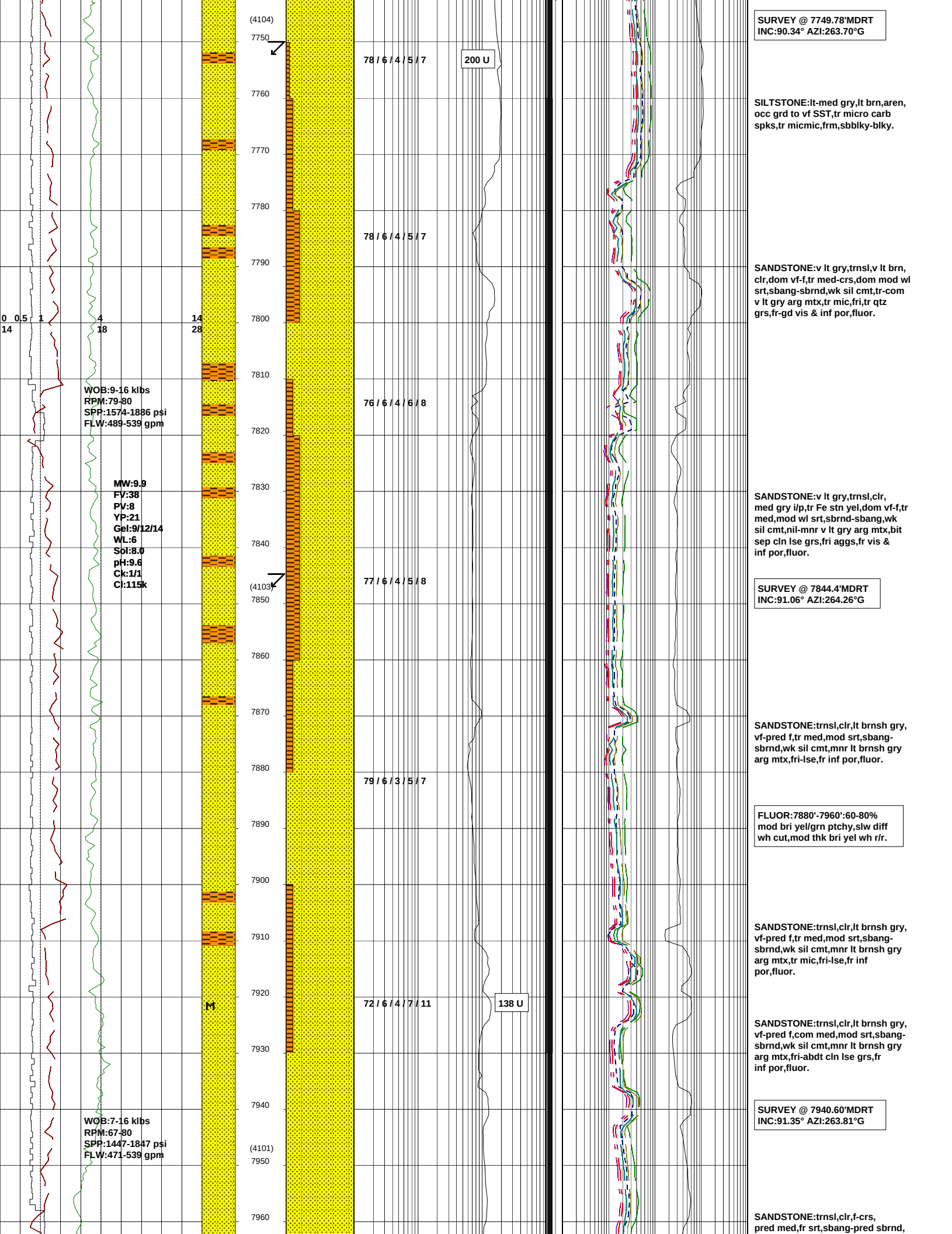


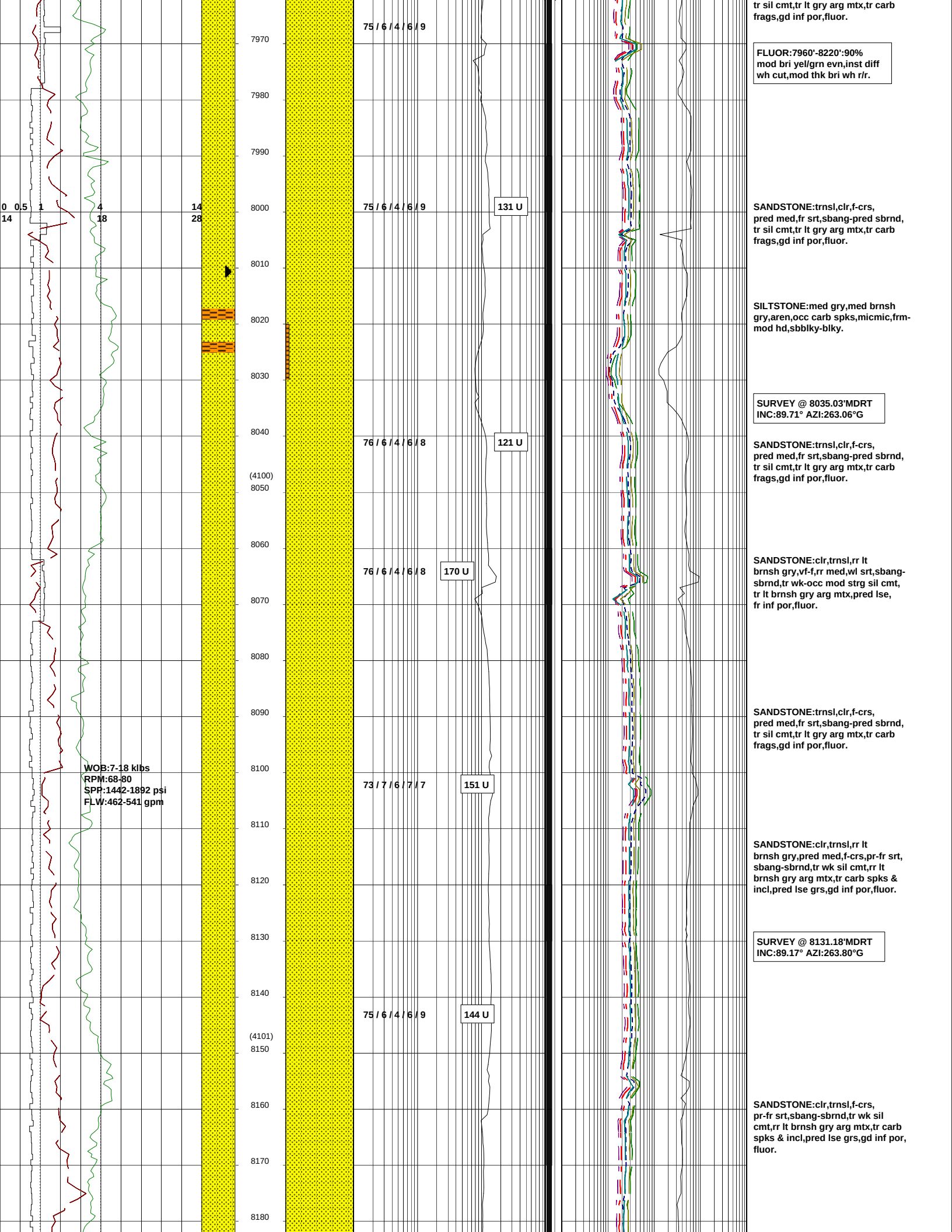


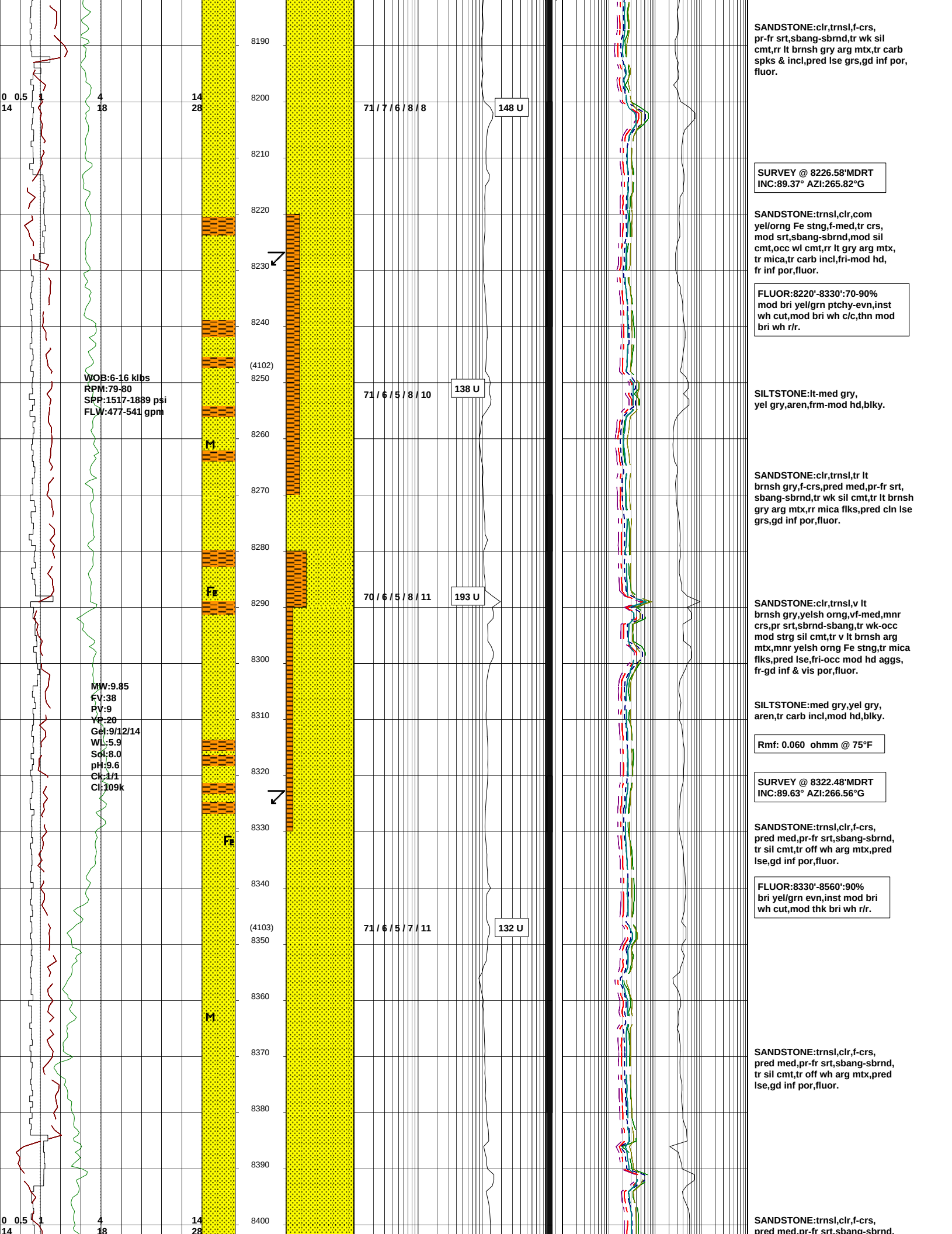


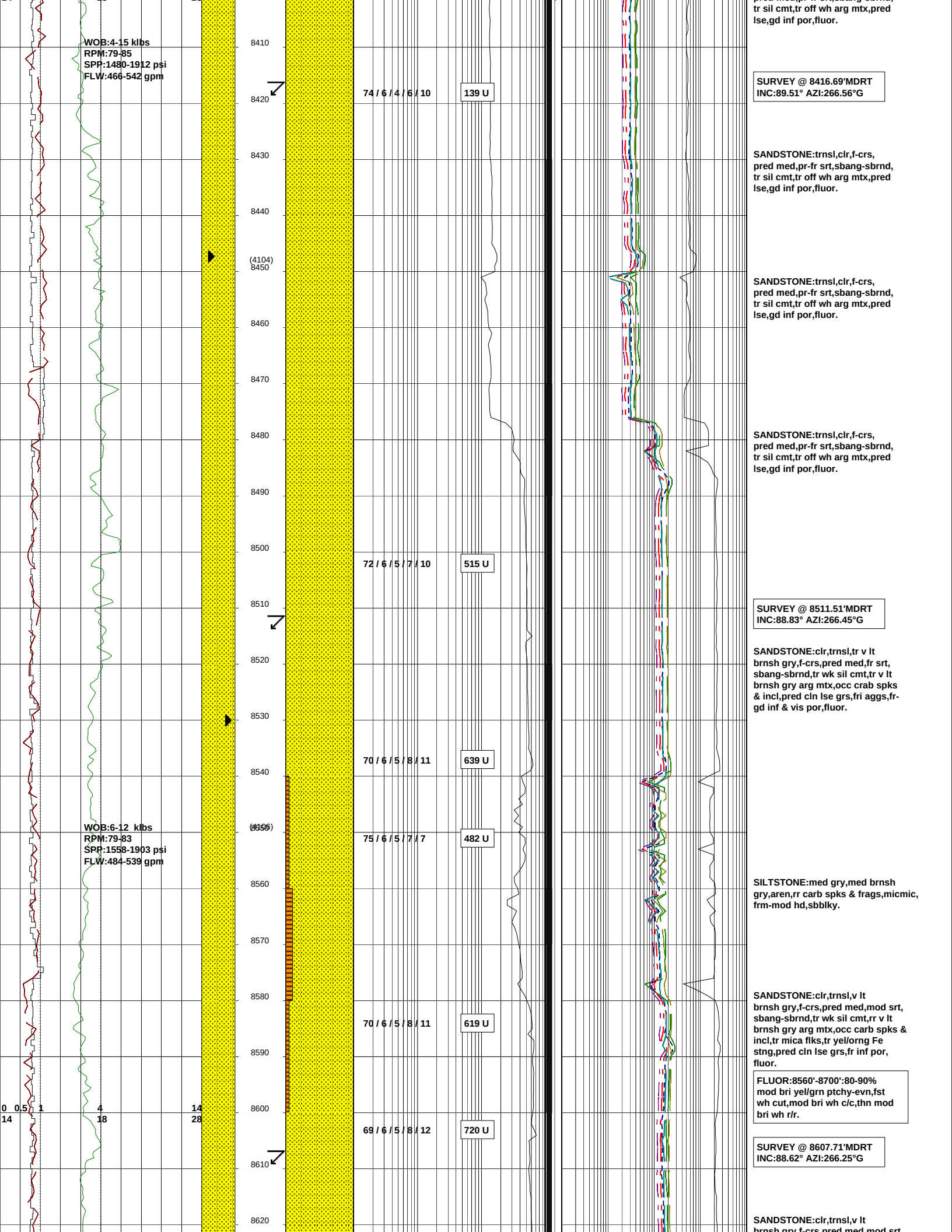


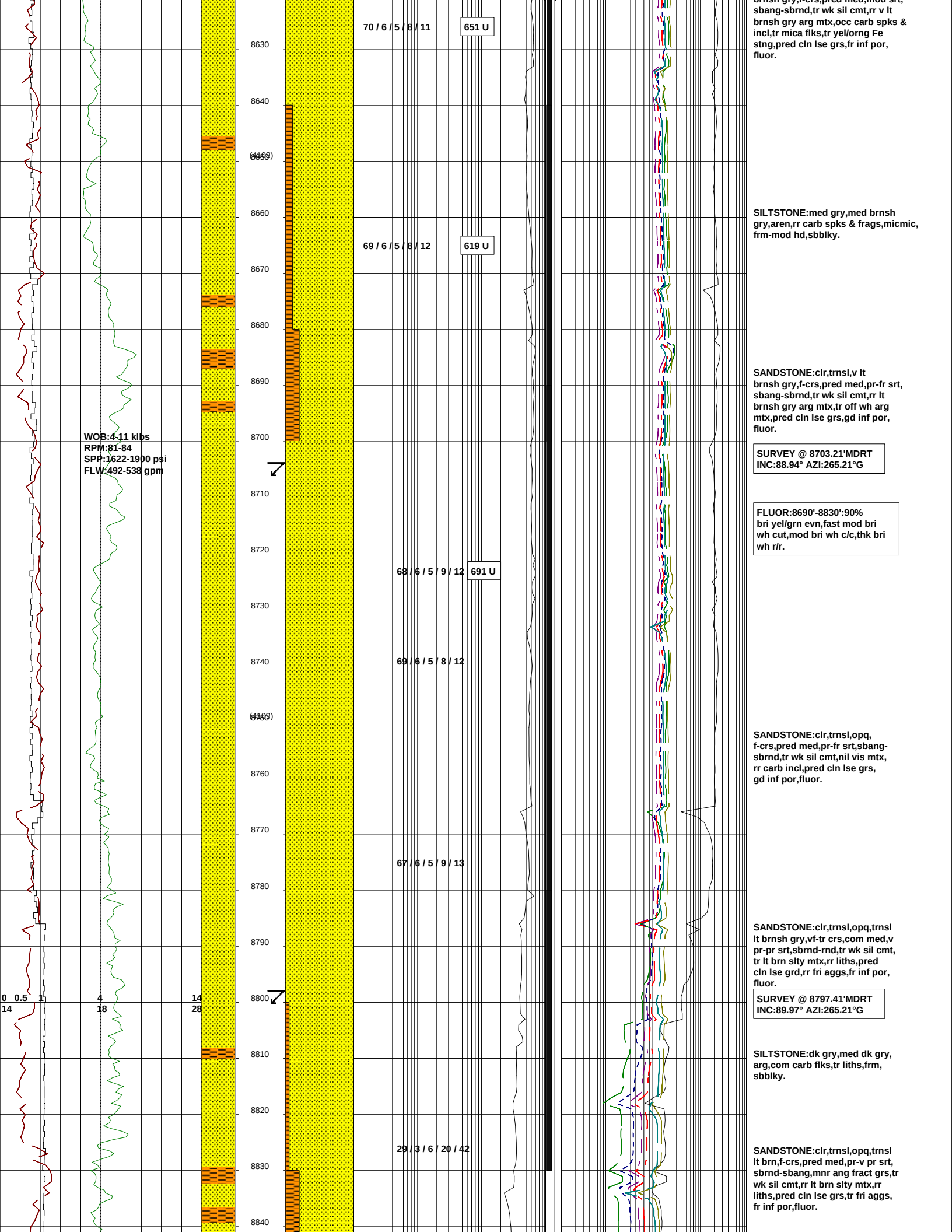


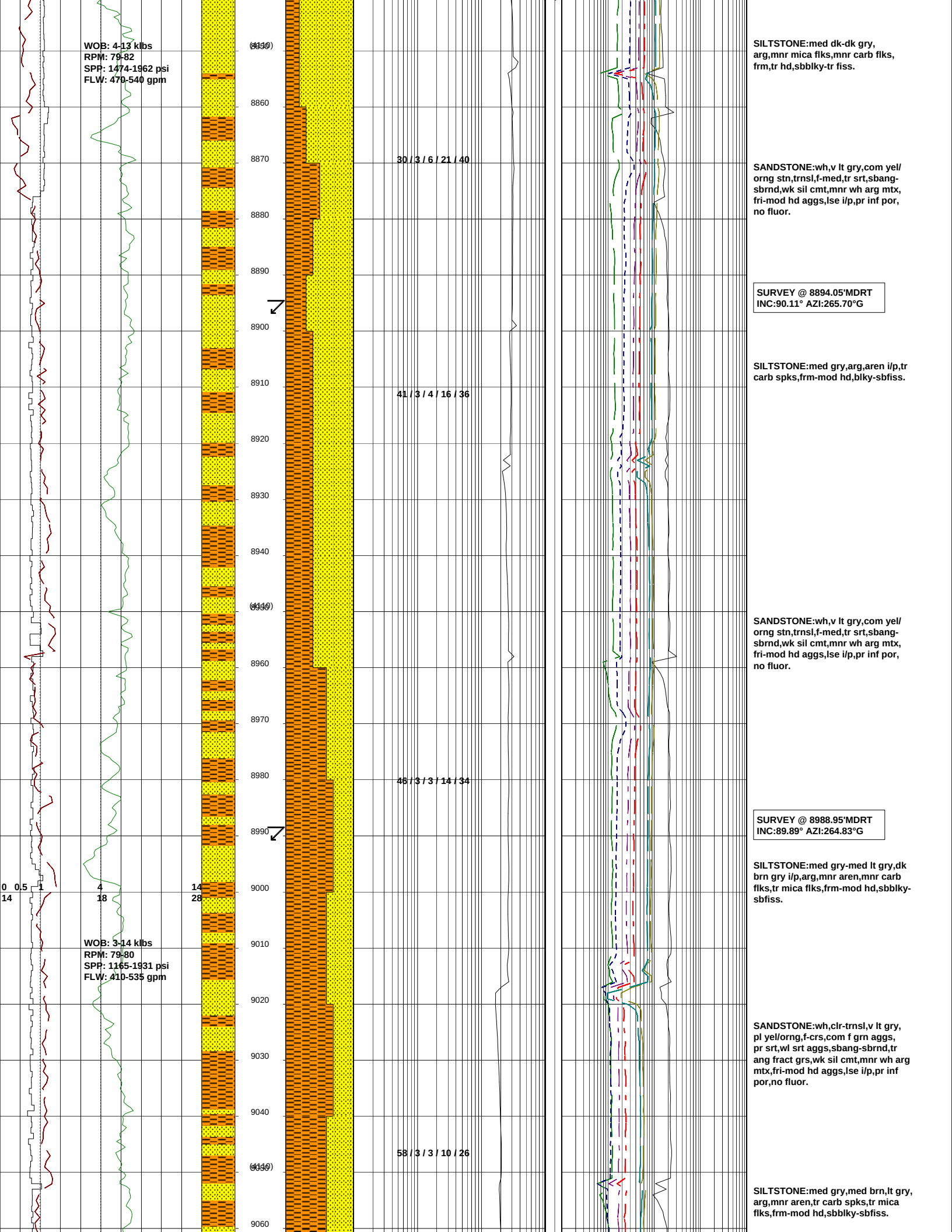


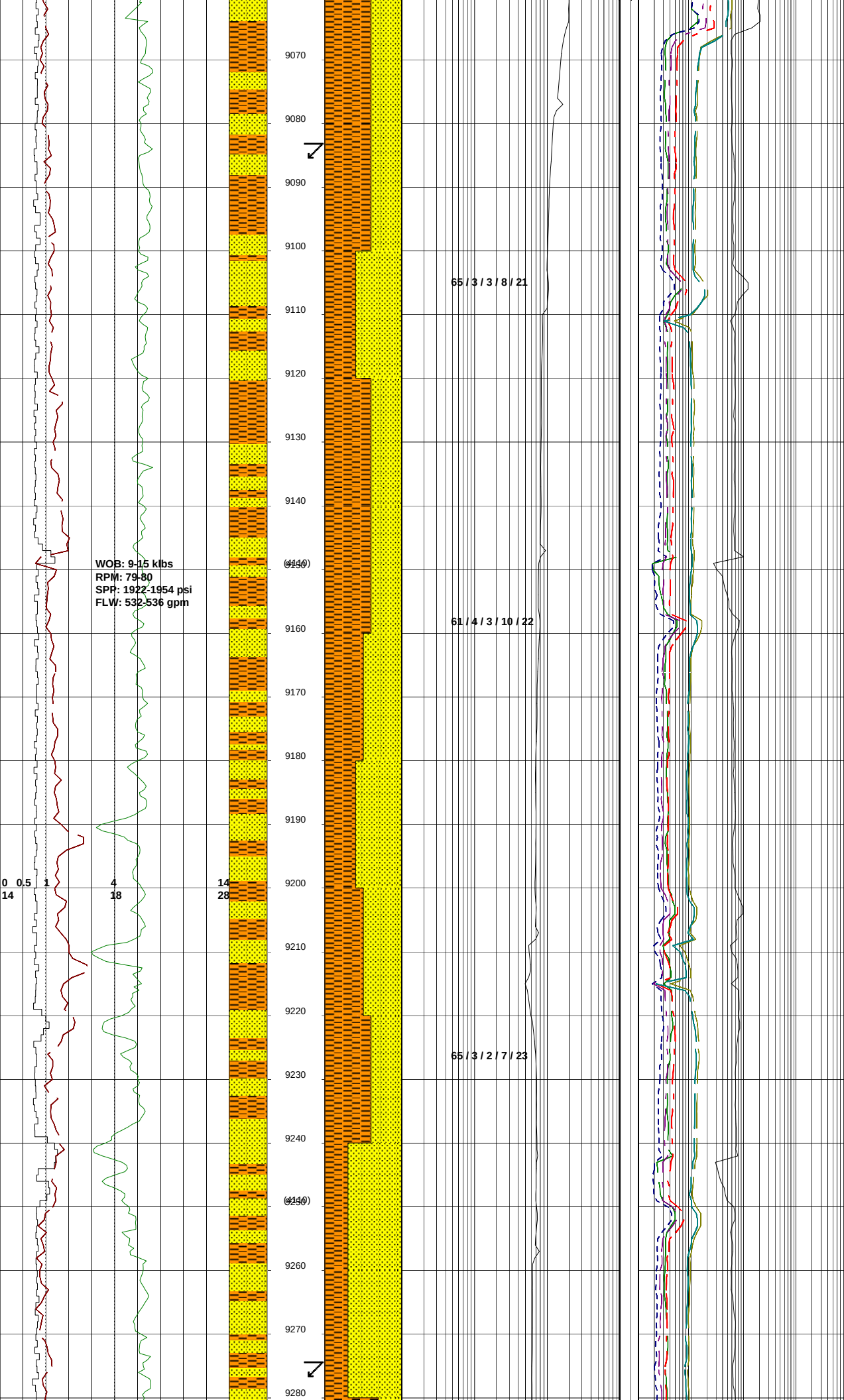












SANDSTONE:trnsI,clr,lt gry,
med brn,com yell/orng Fe stn,f-crs,
pred med,pr-fr srt,sbang-sbrnd,wk
sil cmt,mnr lt yel brn arg mtx,tr
qtz ovghts,fri-mod hd aggs,com diss
agg qtz grs,pr inf por,no fluor.

SURVEY @ 9083.05'MDRT
INC:90.09° AZI:264.72°G

SILTSTONE:med gry,arg,occ aren,
rr carb spks,tr micmic,mod hd,blky-
sbbiky.

SANDSTONE:trnsI,clr,lt gry,
med brn,com yell/orng Fe stn,f-crs,
pred med,pr-srt,sbang-sbrnd,wk
sil cmt,mnr lt yel brn arg mtx,tr qtz
ovghts,fri-mod hd aggs,com diss
agg qtz grs,pr inf por,no fluor.

SILTSTONE:med gry,tr lt gry,arg,
mnr aren,tr carb spks,tr liths,
frm,sbfiss-sbbiky.

SURVEY @ 9178.65'MDRT
INC:90.14° AZI:264.64°G

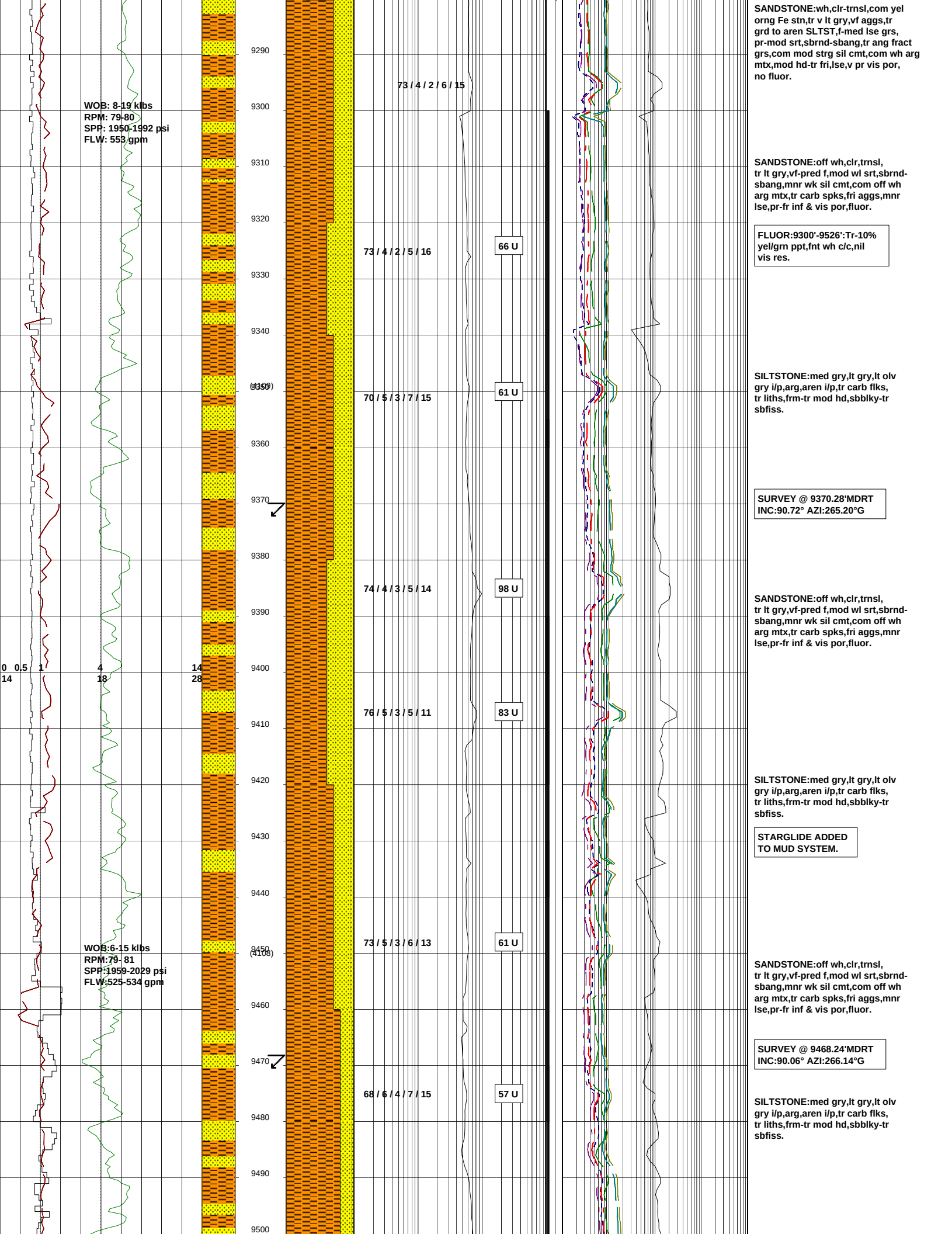
SANDSTONE:wh,trnsI,v lt gry,tr
yell/orng Fe stn,vf-tr f,mnr f-crs
lse grs,pr-mod srt,sbrnd-rnd,tr ang,
com mod sil cmt,com wh arg mtx,
mod hd-tr fri,lse,ti vis por,no fluor.

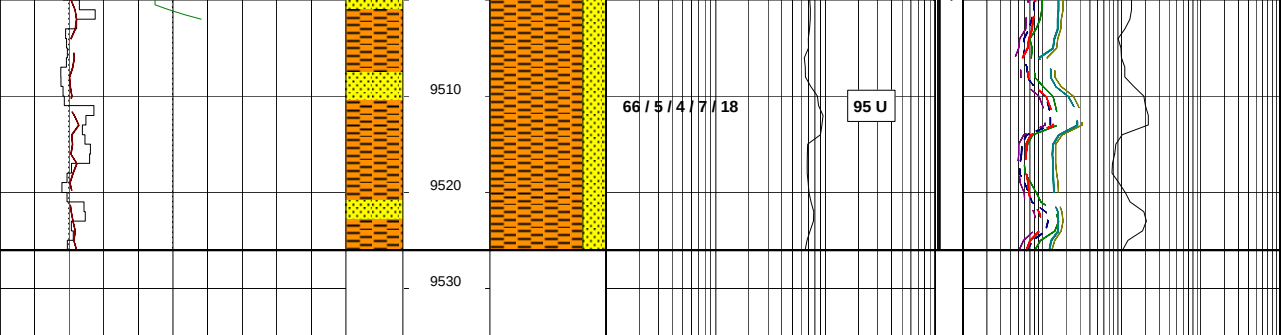
SILTSTONE:med gry,lt gry,lt olv
gry i/p,arg,aren i/p,tr carb flks,
tr liths,frm-tr mod hd,sbbiky-tr
sbfiss.

SANDSTONE:wh,trnsI,v lt gry,com
yell/orng Fe stn,vf-tr f,mnr f-crs
lse grs,pr-mod srt,sbrnd-rnd,tr ang,
com mod sil cmt,com wh arg mtx,
mod hd-tr fri,lse,ti vis por,no fluor.

SILTSTONE:med gry,lt gry,lt olv
gry i/p,arg,aren i/p,tr carb flks,
tr liths,frm-tr mod hd,sbbiky-tr
sbfiss.

SURVEY @ 9274.33'MDRT
INC:90.03° AZI:264.53°G





SANDSTONE:off wh,clr,trnsI,
tr lt gry,vf-pred f,mod wl srt,sbrnd-
sbang,mnr wk sil cmt,com off wh
arg mtx,tr carb spks,fri aggs,mnr
lse,pr-fr inf & vis por,fluor.

MCKINLAY 12 REACHED TD
@ 15:45 HRS ON 17-11-2017